

Introduction to Manifolds

Labix

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Abstract

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1 Introduction to Manifolds

1.1 Pseudogroup of Transformations

Definition 1.1.1 (The Set of Homeomorphisms of Open Subsets)

Let $(X, \mathcal{T})$ be a topological space. Define the set of homeomorphisms of X between open subsets to be

$$H(X) = \{f : U \rightarrow V \mid U, V \in \mathcal{T}, f \text{ is a homeomorphism}\}$$

We establish the definition of manifolds in a slightly non-standard way via the use of pseudogroups.

Definition 1.1.2 (Pseudogroup of Transformations)

Let $(X, \mathcal{T})$ be a topological space. A pseudogroup of transformations on X is a subset of homeomorphisms

$$\Gamma \subseteq H(X)$$

such that the following are true.

- (Cover) There exists an indexing set I and $f_i : U_i \rightarrow V_i \in \Gamma$ for $i \in I$ such that $X = \bigcup_{i \in I} U_i$.
- (Restriction) If $f : U \rightarrow V \in \Gamma$ and $W \subseteq U$, then $f|_W \in \Gamma$.
- (Local) If $f : U \rightarrow V \in H(X)$ and there exists an open cover $U = \bigcup_{i \in I} U_i$ such that $f|_{U_i} \in \Gamma$, then $f \in \Gamma$.
- (Inverse) If $f \in \Gamma$ then $f^{-1} \in \Gamma$.
- (Composition) If $f_1 : U_1 \rightarrow V_1 \in \Gamma$ and $f_2 : V_1 \rightarrow W_2 \in \Gamma$ such that $V_1 \cap V_2 \neq \emptyset$, then

$$f_2 \circ f_1 : f_1^{-1}(V_1 \cap V_2) \rightarrow f_2(V_1 \cap V_2)$$

lies in Γ

Manifolds is more or less some sort of a surface living in $\mathbb{R}^n$. To translate notions of differentiability to a manifold we want to cover the manifold with open sets isomorphic to $\mathbb{R}^m$. The pseudogroup of transformation is essentially determining what structure we want to investigate.

Denote Γ_{C^r} the pseudogroup where all transformations are C^r . Different values of r encode different information on the manifold since the underlying transformations between the open sets that cover the manifold change.

Definition 1.1.3 (C^r -Differentiable Maps) Let $U, V \subseteq \mathbb{R}^n$ be open. Denote $C^r(U, V)$ the set of all r times differentiable maps from U to V such that the r th derivative is continuous. By convention, $C^0(U, V)$ is the set of all continuous maps from U to V and $C^k(\mathbb{R}^n) = C^k(\mathbb{R}^n, \mathbb{R}^n)$. $C^\infty(U, V)$ denotes the set of all infinitely differentiable (smooth) maps from U to V .

Example 1.1.4

The following are pseudogroups of transformations.

- $\Gamma(\mathbb{R}^n, C^0) = H(\mathbb{R}^n)$.
- $\Gamma(\mathbb{R}^n, C^k) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f \in C^k(U, V) \text{ and } f^{-1} \in C^k(V, U)\} \text{ for } 1 \leq k \leq \infty$.
- $\Gamma(\mathbb{R}^n, C^\infty) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f \in C^\infty(U, V) \text{ and } f^{-1} \in C^\infty(V, U)\}$.
- $\Gamma(\mathbb{R}^n, \mathcal{A}) = \{f : U \rightarrow V \in H(\mathbb{R}^n) \mid f, f^{-1} \text{ are analytic}\}$.
- $\Gamma(H^n, C^\infty) = \{f : U \rightarrow V \in H(H^n) \mid f \in C^\infty(U, V) \text{ and } f^{-1} \in C^\infty(V, U)\}$.

1.2 Manifolds and their Structures

Definition 1.2.1 (Atlas)

Let M be a topological space. An atlas of M compatible with a pseudogroup of transformations Γ is a family of pairs $\{(U_i, \phi_i) \mid i \in I\}$, called charts, such that the following are true.

- $\{U_i \mid i \in I\}$ is an open cover of M .
- Each ϕ_i is a homeomorphism of U_i onto an open set of S .
- Whenever $U_i \cap U_j$ is nonempty, the mapping $\phi_j \circ \phi_i^{-1}$ of $\phi_i(U_i \cap U_j)$ onto $\phi_j(U_i \cap U_j)$ is an element of Γ .

This definition complements well with our explanation. For an atlas to make sense with a pseudogroup, the target (range) should be the underlying $\mathbb{R}^m$. The U_i should at least cover M and be part of M . And finally areas with overlapping should have maps that allow transition between two covers, which is what the third condition is saying.

Definition 1.2.2 (Compatible Charts) Let (U, ϕ) be a chart of a space M . We say that it is compatible with another chart (V, φ) if $\varphi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \varphi(U \cap V)$ is in Γ . We say that (U, ϕ) is compatible with an atlas $\{(U_i, \phi_i) \mid i \in I\}$ of M if (U, ϕ) is compatible with every chart in the atlas.

To prove that two atlases A and A' are not equal, one should find a chart in A that is incompatible with a chart in A' , or vice versa.

Definition 1.2.3 (Complete Atlas) A complete atlas on a topological space M compatible with Γ is a atlas of M compatible with Γ which is not contained in any other atlas M compatible with Γ .

Complete atlases are also called maximal atlases.

Lemma 1.2.4 Every atlas of M compatible with Γ is contained in a unique complete atlas.

Proof Let $A = \{(U_i, \phi_i)\}$ be an atlas of M compatible with Γ . Let $\tilde{A}$ be the family of all pairs (U_i, ϕ_i) such that $\phi : U_i \subseteq M \rightarrow S$ is a homeomorphism such that $\phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$ is an element of Γ whenever $U_i \cap U_j$ is nonempty. Then $\tilde{A}$ is the complete atlas of M containing A . ■

However note that this is not to say that M has a unique Γ -structure.

By collecting atlas and using the complete atlas of a topological space, we can talk about all possible transitions between the open covers.

Definition 1.2.5 (Manifolds)

Let Γ be a pseudogroup of transformations. A Γ -manifold is a Hausdorff topological space M that is second countable, together with a fixed complete atlas compatible with Γ .

The pseudogroup of transformation gives a local chart around every point in the manifold so that the manifold is locally Euclidean. This means that nearby every point, the manifold looks like Euclidean space.

Definition 1.2.6 (Common Types of Manifolds)

Let Γ be a pseudogroup of transformations. Let M be a Γ -manifold.

- We say that M is a topological manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^0)$.
- Let $1 \leq k < \infty$. We say that M is a C^k -differentiable manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^k)$.
- We say that M is a smooth manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{C}^\infty)$.
- We say that M is a real analytic manifold if $\Gamma = \Gamma(\mathbb{R}^n, \mathcal{A})$.
- We say that M is a smooth manifold with boundary if $\Gamma = \Gamma(H^n, \mathcal{C}^\infty)$.

Notice that it is not possible for a manifold to locally look like both $\mathbb{R}^n$ and $\mathbb{R}^m$. By invariance of domain, if two open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively are homeomorphic, then $n = m$.

Lemma 1.2.7 The following are true regarding the different types of manifolds.

- Every smooth manifold is a differentiable manifold of class C^r for any $r \in \mathbb{N}$
- Every differentiable manifold of class C^r is a differentiable manifold of class C^{r-1} for $r \geq 1$
- Every differentiable manifold of class C^r is a topological manifold

A chart on a real manifold M of dimension n is actually an open set together with n coordinate functions.

Indeed if (U, ϕ) is a chart of M , then $\phi : U \rightarrow \mathbb{R}^n$ is a map to $\mathbb{R}^n$. If we are interested in the particular components of ϕ , we write $\phi = (x^1, \dots, x^n)$ where $x^i : U \rightarrow \mathbb{R}$ is the coordinate function sending a point on U to its i th coordinate. All the x^i 's combine to recover ϕ . If we would like to talk about the actual coordinate on $\mathbb{R}^n$ instead of the coordinate function, we write x_i instead.

Definition 1.2.8 (Transition Functions)

Let M be a manifold. Let $\{(U_i, \phi_i) \mid i \in I\}$ be an atlas on M . The transition functions on M are the functions

$$\phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$$

for $U_i \cap U_j \neq \emptyset$.

1.3 Manifolds with Boundaries

Definition 1.3.1 (Boundary Charts)

Let M be a smooth manifold of dimension n with boundary. Let $p \in \partial M$. Let (U, ϕ) be a chart at p . We say that (U, ϕ) is a boundary chart at p if the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 1.3.2 (The Boundary of a Manifold)

Let M be a smooth manifold of dimension n with boundary. Define the boundary of M to be

$$\partial M = \{p \in M \mid p \text{ admits a boundary chart}\}$$

Proposition 1.3.3

Let M be a smooth manifold with boundary of dimension n . Then ∂M is a smooth manifold of dimension $n - 1$ with atlas

$$\{(U_\alpha|_{\partial M}, \phi_\alpha|_{\partial M})\}$$

for $\{(U_\alpha, \phi_\alpha)\}$ the atlas on M .

1.4 Maps between Manifolds

Definition 1.4.1 (Morphism of Manifolds)

Let Γ be a pseudogroup of transformations. Let M, N be Γ -manifolds whose atlases are given by $\{(U_i, \phi_i) \mid i \in I\}$ and $\{(V_j, \psi_j) \mid j \in J\}$ respectively. Let $f : M \rightarrow N$ be a function. We say that f is a morphism if

$$\psi_j \circ f \circ \phi_i^{-1} \in \Gamma$$

whenever $f(U_i) \cap V_j \neq \emptyset$.

Lemma 1.4.2

Let Γ be a pseudogroup of transformations. Let M, N, K be Γ -manifolds. Let $f : M \rightarrow N$ and $g : N \rightarrow K$ be morphisms. Then $g \circ f$ is a morphism.

Lemma 1.4.3

Let Γ be a pseudogroup of transformations. Let M be a Γ -manifold. Then $\text{id}_M : M \rightarrow M$ is a morphism.

Definition 1.4.4 (Isomorphisms)

Let Γ be a pseudogroup of transformations. Let M, N be Γ -manifolds. Let $f : M \rightarrow N$ be a morphism. We say that f is an isomorphism if there exists a morphism $g : N \rightarrow M$ such that

$$g \circ f = \text{id}_M \quad \text{and} \quad f \circ g = \text{id}_N$$

1.5 Comparing Different Structures on Manifolds

So far, we have seen no less than 4 structures on manifolds. In other words, we have at least 4 different classes of manifolds.

- Topological manifolds with C^0 structure
- C^k -differentiable manifolds for $k \geq 1$ with C^k -structure
- Smooth manifolds with C^∞ -structure
- Real analytic manifolds with analytic structure

There is another important class of manifolds which are complex analytic (holomorphic) manifolds, but we will only discuss real manifolds in these notes. What are the relations between all such structures? This section will answer that.

Proposition 1.5.1

Let $k \geq 1$. Let M be a C^k manifold with atlas $\mathcal{A}$. Then $\mathcal{A}$ is compatible with a unique C^∞ atlas.

ref: hirsch differential topology thm2.9

Proposition 1.5.2

Let $k \geq 1$. Let M, N be smooth manifolds. Then M and N are diffeomorphic if and only if M and N are C^k -diffeomorphic.

Proposition 1.5.3

The following is a complete list of topological (smooth) manifolds of dimension 1 up to homeomorphism.

- $\mathbb{R}$.
- S^1 .

Theorem 1.5.4 ((Moise's theorem))

Let M be a topological manifold of dimension 2 or 3. Then M admits a unique smooth structure.

Proposition 1.5.5

Let M, N be topological manifolds of dimension both 2 or 3. Then M and N are homeomorphic if and only if M and N are diffeomorphic.

2 New Smooth Manifolds from Old

2.1 Subsets of a Manifold

Let M be a smooth manifold. We may ask the conditions for a subset S of M to be a smooth manifold itself, whose smooth structure comes inherited from that of M . When S is open, the answer is simple.

Proposition 2.1.1 Let M be a smooth manifold. Let $N \subseteq M$ be an open subset. Then N is a smooth manifold and $\dim(N) = \dim(M)$.

Proof It is easy to see that N is second countable and Hausdorff. Moreover, if $\{(U_i, \phi_i) \mid i \in I\}$ is a complete atlas of M , then it is easy to see that $\{(U_i \cap N, \phi_i|_{U_i \cap N})\}$ is a complete atlas of N . ■

Let M be a smooth manifold. Even when $S \subseteq M$ is not open, there are still ways for S to be smooth manifold itself that inherits the smooth structure from M . This motivates the following definition.

Definition 2.1.2 (Embedded Submanifolds) Let M be a smooth manifold of dimension m . We say that $N \subseteq M$ is a submanifold of dimension n if for every $p \in N$ there exists a chart (U, ϕ) on M with $p \in U$ such that

$$\phi(U \cap N) = \{(x_1, \dots, x_m) \in \phi(U) \mid m - n \text{ out of the } m \text{ coordinate functions vanish}\}$$

In this case we say that Y has codimension $m - n$.

By renumbering the coordinate functions, the condition translates to

$$\phi(U \cap N) = \{(x_1, \dots, x_m) \in \phi(U) \mid x_{n+1} = x_{n+2} = \dots = x_m = 0\}$$

Embedded submanifolds are also called regular submanifolds.

We need to show that such a definition indeed gives a smooth manifold in its own right.

Proposition 2.1.3 Let M be a smooth manifold with atlas $\{(U_i, \phi_i) \mid i \in I\}$. Let $Y \subseteq M$ be an embedded submanifold of M . Then Y is also a smooth manifold in its own right with atlas

$$\{(U_i \cap Y, \phi|_{U_i \cap Y}) \mid i \in I\}$$

Proof It is clear that the given set covers Y . Each of them are also charts since $\phi|_{U_i \cap Y} : U_i \cap Y \rightarrow \mathbb{R}^n$ is a diffeomorphism since it is the restriction of a diffeomorphism. Assume that two charts on M $(U, \phi = (x^1, \dots, x^m))$ and $(V, \psi = (y^1, \dots, y^m))$ intersect. Relabel ϕ and ψ so that on U and V , the last $n - k$ coordinates vanish on Y . Then for $p \in U \cap V \cap Y$, we have $\phi|_{U \cap Y}(p) = (x^1, \dots, x^k)$ and $\psi|_{V \cap Y}(p) = (y^1, \dots, y^k)$. Therefore

$$\psi|_{V \cap Y} \circ (\phi|_{U \cap Y})^{-1}(x^1, \dots, x^k) = (y^1, \dots, y^k)$$

$\psi \circ \phi^{-1}$ is C^∞ so $y^1, \dots, y^k$ each are C^∞ functions of $x^1, \dots, x^k$ so that $\psi|_{V \cap Y} \circ (\phi|_{U \cap Y})^{-1}$ is C^∞ . Hence any two charts are compatible on Y . ■

Below is a criterion to check whether charts of a subset of a manifold satisfy the condition for Y to be a submanifold.

Proposition 2.1.4 Let $f : M \rightarrow \mathbb{R}^n$ be a smooth map where M is an m dimensional manifold, with components $f = (f_1, \dots, f_n)$. Let $p \in f^{-1}(0)$ be arbitrary. Let (U, ϕ) be any chart of M at p .

If the following $n \times m$ matrix

$$\begin{pmatrix} \frac{\partial(f_1 \circ \phi^{-1})}{\partial x_1} \big|_{\phi(p)} & \cdots & \frac{\partial(f_1 \circ \phi^{-1})}{\partial x_m} \big|_{\phi(p)} \\ \vdots & \ddots & \vdots \\ \frac{\partial(f_n \circ \phi^{-1})}{\partial x_1} \big|_{\phi(p)} & \cdots & \frac{\partial(f_n \circ \phi^{-1})}{\partial x_m} \big|_{\phi(p)} \end{pmatrix}$$

has rank n (regardless of choice of chart), then $f^{-1}(0)$ is a submanifold of M of dimension $m - n$.

Proof Notation change: Write $\frac{\partial f_i}{\partial x^j}$ in place of $\frac{\partial(f_i \circ \phi^{-1})}{\partial x_j}$.

Since the given matrix M has rank n , we can relabel the f_i and the x_j so that the first $n \times n$ block matrix $\left(\frac{\partial f_i}{\partial x^j}\right)_{1 \leq i, j \leq n}$ has non-zero determinant. In other words, our $m \times n$ matrix is now of the form

$$\left(\left(\frac{\partial f_i}{\partial x^j} \right)_{1 \leq i, j \leq n} \quad * \right) \in \text{Mat}_{m \times n}$$

Now notice that $Y = f^{-1}(0)$ is our set in question. Let $p \in Y$. Let $(U, \phi = (x^1, \dots, x^m))$ be an arbitrary choice of chart at p on M . Then notice that $(U, (f_1, \dots, f_n, x^{n+1}, \dots, x^m))$ is also a chart on M . Indeed $f_1, \dots, f_n$ are assumed to be smooth so that it is compatible with the atlas of M . Now the Jacobian of ϕ^{-1} composed of this chart, which is an $m \times m$ matrix, is given by the formula

$$\begin{pmatrix} \left(\frac{\partial f_i}{\partial x^j} \right)_{1 \leq i, j \leq n} & * \\ 0 & I \end{pmatrix} \in \text{Mat}_{m \times m}$$

Its determinant is precisely

$$\left| \left(\frac{\partial f_i}{\partial x^j} \right)_{1 \leq i, j \leq n} \right|$$

which is non-zero by assumption. Then the inverse function theorem applies.

In the chart $(U, (f_1, \dots, f_n, x^{n+1}, \dots, x^m))$, the set $Y = f^{-1}(0)$ is obtained by setting the vanishing of the first m coordinate functions $f_1, \dots, f_n$. Thus $\phi(U \cap Y)$ is now equal to $\phi(U) \cap \mathbb{R}^{m-n}$ so that Y is an embedded submanifold. ■

For example, consider the equation $y^2 = x^2 - x$. Consider the set $\{(x, y) \in \mathbb{R}^2 \mid y^2 - x^2 + x = 0\}$. To show that this is a submanifold of $\mathbb{R}^2$, we use the above theorem by setting $f(x, y) = y^2 - x^2 + x$. Then our set becomes

$$f^{-1}(0) = \{(x, y) \in \mathbb{R}^2 \mid y^2 - x^2 + x = 0\}$$

The gradient of f is $(-2x + 1, 2y)$, which vanishes only at $(1/2, 0)$, outside of the curve. Thus by the above theorem, $y^2 = x^2 - x$ is a submanifold.

Note that this is a slightly weaker statement of the preimage theorem 4.3.3. In that theorem, instead of using the C^∞ map $F = (f_1, \dots, f_n)$ that maps to $\mathbb{R}^n$, it considers in what situation will a smooth map of manifolds $f : M \rightarrow N$ give a submanifolds on a level set $f^{-1}(c)$ for $c \in N$.

2.2 Quotient Manifolds

3 Basic Properties of Smooth Manifolds

3.1 Partitions of Unity

Smooth partitions of unity is unique to manifold that are locally $\mathbb{R}^n$. Complex manifolds would have a more rigid structure in the sense that they do not have a partition of unity. In our case of smooth manifolds, we can construct a smooth partition of unity. This allows us to piece together functions defined on charts on a manifold.

Definition 3.1.1 (Support of a Function) Let $f : M \rightarrow \mathbb{R}$ be a function on a manifold M . Define the support of f to be

$$\text{supp}(f) = \overline{\{p \in M | f(p) \neq 0\}}$$

Definition 3.1.2 (Compact Exhaustion) Let X be a topological space. A compact exhaustion of X is a sequence of open subsets $\{K_i | i \in \mathbb{N}\}$ such that $K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots$ such that each $\overline{K_i}$ is compact, $\overline{K_i} \subset K_{i+1}$ and $X = \bigcup_{i=1}^{\infty} K_i$.

Proposition 3.1.3 Every manifold M admits a compact exhaustion.

Proof Firstly, since M has a countable basis from second countability, and M is locally Euclidean, we can find a countable basis for the topology of M , namely $\{V_j | j \in \mathbb{N}\}$ such that each V_j has compact closure. Indeed, we can just throw out any V_j from the original list that does not have compact closure. Set $G_1 = V_1$. Suppose inductively that $G_k = \bigcup_{i=1}^{j_k} V_i$. Let j_{k+1} be the smallest positive integer greater than j_k such that $\overline{G_k} \subset \bigcup_{i=1}^{j_{k+1}} V_i$ and put $G_{k+1} = \bigcup_{i=1}^{j_{k+1}} V_i$. This gives a compact exhaustion as required. ■

Definition 3.1.4 (Partitions of Unity) A partition of unity on a manifold M is a collection of smooth non-negative functions $\{\theta_i : M \rightarrow \mathbb{R} | i \in I\}$ such that the following are true.

- For all $p \in M$ and $i \in I$, $0 \leq \theta_i(p) \leq 1$
- For each $p \in M$, there is a neighbourhood U of p such that there is only a finite number of θ_i with non-empty support.
- $\sum_{i \in I} \theta_i = 1$

Definition 3.1.5 (Subordinate Partitions) We say that a partition of unity $\{\theta_i : M \rightarrow \mathbb{R} | i \in I\}$ for a manifold M is subordinate to an open cover $\{U_\alpha | \alpha \in A\}$ of M if $\text{supp}(\theta_i) \subset U_\alpha(i)$ for every $i \in I$.

Theorem 3.1.6 (Existence of a Partition of Unity) Let M be a manifold and $\{U_\alpha | \alpha \in A\}$ an open cover of M . Then there exists a partition of unity subordinate to the open cover $\{U_\alpha | \alpha \in A\}$.

Proof We first prove a result in $\mathbb{R}^n$. Namely: Let $C(r) = \{(x_1, \dots, x_n) \in \mathbb{R}^n | |x_i| < r\}$ be the open cube of side length $2r$ centered at the origin. Then there exists a non-negative smooth function φ on $\mathbb{R}^n$ such that $\varphi(x) = 1$ for $x \in \overline{C(1)}$ and $\varphi(x) = 0$ for $x \in \mathbb{R}^n \setminus C(2)$. We just have to do it in dimension $n = 1$ since we can just take product of such functions. One possible way to write down a formula is: Start with

$$f(t) = \begin{cases} e^{-1/t} & \text{if } t > 0 \\ 0 & \text{if } t \leq 0 \end{cases}$$

Put $g(t) = \frac{f(t)}{f(1)+f(1-t)}$ which is non-negative, smooth and equal to 1 for $t \geq 1$ and 0 for $t \leq 0$. Then setting $\varphi(t) = g(t+2)g(2-t)$ gives the function as required.

Now for the main proof. Construct a compact exhaustion for M and set $G_0 = \emptyset$. For a point $p \in M$, let i_p be the largest integer such that $p \in M \setminus \overline{G_{i_p}}$. We also pick an index $\alpha_p \in A$ such that $p \in U_{\alpha_p}$. Now let (V, τ) be a chart around p with $\tau(p) = 0$ and such that

$$V \subset U_{\alpha_p} \cap (G_{i_p+2} \setminus \overline{G_{i_p}})$$

and such that $\tau(V)$ contains the closed cube $\overline{C(2)}$. Now define

$$\psi_p(q) = \begin{cases} \varphi \circ \tau(q) & \text{if } q \in V \\ 0 & \text{otherwise} \end{cases}$$

where φ is the smooth function constructed above. Clearly ψ_p is a smooth function on M taking the value 1 on some open neighbourhood W_p of p with compact support contained in $V \subset U_{\alpha_p} \cap (G_{i_p+2} \setminus \overline{G_{i_p}})$.

For each $i \geq 1$, choose a finite set of points $p \in M$ whose neighbourhoods W_p as above cover $\overline{G_i} \setminus G_{i-1}$. This is possible since the $\overline{G_i} \setminus G_{i-1}$ is compact. Order all functions ψ_p corresponding to such points p in a sequence

$$\psi_1, \psi_2, \psi_3, \dots$$

Clearly, for every point in M , only finitely many of the ψ_j are nonzero at that point since the set of ψ with support contained in each G_i is finite by construction. This means that

$$\psi = \sum_{j=1}^{\infty} \psi_j$$

is a well defined smooth function on M with $\psi(p) > 0$ everywhere. Then define

$$\theta_i = \frac{\psi_i}{\psi}$$

Then these θ_i have all the required properties. ■

This tool will be useful for us to patch up local expression and normalize it so that the output is a globally defined object on the manifold, provided that they agree on intersections.

3.2 The Whitney Embedding Theorem

Theorem 3.2.1 (The (Strong) Whitney Embedding Theorem) Any smooth manifold of dimension m can be smoothly embedded into $\mathbb{R}^{2m}$.

4 The Tangent Space

4.1 Tangent Space

Conventionally there are two ways of defining a tangent vector on a smooth manifold. We will take the more geometric approach by considering curves on the manifold. To begin, we need some algebra.

Definition 4.1.1 ($\mathbb{R}$ -Algebra of Germs of $\mathcal{C}^\infty$ Functions) Let M be a manifold. Let $p \in M$. Define the $\mathbb{R}$ -Algebra of Germs of Functions

$$\mathcal{C}_{M,p}^\infty = \{(f : U \rightarrow \mathbb{R}, U) \mid p \in U, f \in \mathcal{C}^\infty\} / \sim$$

where we say that $(f, U) \sim (g, V)$ if there exists some open neighbourhood W of p so that $W \subset U \cap V$ and $f|_W = g|_W$.

Since we are only concerned with tangents of the curves in order to form the tangent space, whenever two functions coincide on some open subset, they will have the same tangents there. So indeed the definition makes sense. However, two different germs can still have the same tangent vector. This appears for example when comparing the tangent vector at 0 of the function $f(x) = 0$ and $g(x) = x^2$.

Readers with algebraic geometry background will notice that this is very similar to the definition of sheaves. Indeed it is a sheaf of $\mathbb{R}$ -algebra over M . Moreover, these germs are elements of the stalk of a sheaf.

Definition 4.1.2 (Equivalence Class of Curves) Let M be a smooth manifold of dimension n . Let $p \in M$ and (U, ϕ) a coordinate chart on p . We say that two curves $\gamma_1, \gamma_2 : (-1, 1) \rightarrow \mathbb{R}^n$ are equivalent if

$$\left. \frac{d(\phi \circ \gamma_1)(t)}{dt} \right|_0 = \left. \frac{d(\phi \circ \gamma_2)(t)}{dt} \right|_0$$

they have the same derivative at 0 under the coordinate chart.

It is easy to check that this equivalence class is not affected by the choice of the coordinate chart.

Definition 4.1.3 (Tangent Vectors) Let M be a smooth manifold. Let $p \in M$ and $[\gamma]$ an equivalence class of curves on M such that $\gamma(0) = p$. Define the tangent vector to the curve γ at p to be the map $X_{\gamma,p} : \mathcal{C}_{M,p}^\infty \rightarrow \mathbb{R}$ defined by

$$X_{\gamma,p}(f) = \left. \frac{d(f \circ \gamma)(t)}{dt} \right|_{t=0}$$

Notice that this construction shows that tangent vectors are elements of the dual space of $\mathcal{C}_{M,p}^\infty$. This will be called the cotangent space in later sections, and it appears more naturally than tangent space. Moreover, the definition varies with γ instead of anything else. By allowing γ to be different curves, we obtain different tangent vectors. In particular, changing the germ f does not lead to new tangents: it only modifies the height of the output as well as the domain.

It is easy to see that two different representatives of the same equivalence class of curves give the same tangent vector.

What follows is that we will show that these are essentially the same construction, where we have the tangent vectors being an explicit construction.

Definition 4.1.4 (Tangent Space) Let M be a manifold. Define the tangent space to M at p to be the set of all tangent vectors of M at p , denoted $T_p(M)$.

Definition 4.1.5 (Standard Basis Vectors of the Tangent Space With Respect to a Chart)

Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart containing p . Define the tangent vector $\frac{\partial}{\partial x^k} \Big|_p : \mathcal{C}_{M,p}^\infty \rightarrow \mathbb{R}$ by

$$\frac{\partial}{\partial x^k} \Big|_p (f) = \frac{\partial f \circ \phi^{-1}}{\partial x^k} \Big|_{\phi(p)}$$

Proposition 4.1.6 (Standard Basis)

Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart containing p . Then an $\mathbb{R}$ -basis for $T_p(M)$ is given by

$$\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$$

Proof By chain rule applied to the map $(f \circ \phi^{-1}) \circ (\phi \circ \gamma) : \mathbb{R} \rightarrow \mathbb{R}^n$, for any $\gamma : (-1, 1) \rightarrow M$ and $f \in \mathcal{C}_{M,p}^\infty$, we have

$$X_{\gamma,p}(f) = \sum_{k=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial x^k} \Big|_{\phi(p)} \cdot \frac{d\gamma_k(t)}{dt} \Big|_{t=0}$$

where $\gamma_k(t) = x^k$. Rewriting $a_k(p) = \frac{d\gamma_k(t)}{dt} \Big|_{t=0}$, we have that

$$X_{\gamma,p}(f) = \sum_{k=1}^n a_k \frac{\partial(f \circ \phi^{-1})}{\partial x^k} \Big|_{\phi(p)}$$

By allowing f to vary, we obtain $X_{\gamma,p} = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \Big|_p$. It remains to show that they span the vector space and that they are linearly independent.

On one hand, given a linear combination $\sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \Big|_p$, let $(U, \phi = (x^1, \dots, x^n))$ be a chart at p and consider the curve $\gamma : (-1, 1) \rightarrow M$ given on local charts by

$$(\phi \circ \gamma)(t) = \begin{pmatrix} x^1(p) + a_1(p)t \\ \vdots \\ x^n(p) + a_n(p)t \end{pmatrix}$$

Then the tangent vector to the curve γ at p is given by

$$\begin{aligned} X_{\gamma,p}(f) &= \frac{d(f \circ \gamma)(t)}{dt} \Big|_{t=0} \\ &= \left(\frac{\partial f \circ \phi^{-1}}{\partial x^1} \Big|_{\phi(p)} \quad \dots \quad \frac{\partial f \circ \phi^{-1}}{\partial x^n} \Big|_{\phi(p)} \right) \cdot \frac{d(\phi \circ \gamma)(t)}{dt} \Big|_{t=0} \\ &= \sum_{k=1}^n a_k(p) \frac{\partial f \circ \phi^{-1}}{\partial x^k} \Big|_{\phi(p)} \end{aligned}$$

for any choice of $f \in \mathcal{C}_{M,p}^\infty$. On the other hand, each of $\frac{\partial}{\partial x^k} \Big|_p$ are linearly independent because if

$$\sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k} \Big|_p = 0$$

then we have that

$$0 = \sum_{k=1}^n a_k(p) \frac{\partial x^i}{\partial x^k} \Big|_p = a_k(p)$$

so that $\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$ is a linearly independent set. ■

This means that a tangent vector of M has the form

$$X_{\gamma,p} = \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k} \Big|_p \quad \text{and} \quad X_{\gamma,p}(f) = \sum_{k=1}^n a_k(p) \frac{\partial(f \circ \phi^{-1})}{\partial x_k} \Big|_{\phi(p)}$$

where $a_k(p) \in \mathbb{R}$ depends on $[\gamma]$ for each k .

This proposition also shows that the choice of γ becomes encoded in the coefficients of the basis elements $a_1(p), \dots, a_n(p) \in \mathbb{R}$. Thus from now on we omit γ in $X_{\gamma,p}$ and simply write X_p . Later we will see that as we smoothly vary the point p , if each $a_1, \dots, a_n$ varies smoothly with points on M so that $a_1, \dots, a_n \in C_{M,p}^\infty$, then X becomes a vector field.

Corollary 4.1.7

Let M be a smooth manifold. Then we have $\dim_{\mathbb{R}}(T_p(M)) = \dim(M)$.

Given two charts overlapping on a point $p \in M$, how does the derivative of the functions change with respect to changing their coordinate charts? The following proposition shows that once again, the question can be reduced to a problem on multivariable calculus.

Proposition 4.1.8

Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ and $(V, y^1, \dots, y^n)$ be two charts containing p . Then

$$\frac{\partial}{\partial x^k} \Big|_p = \sum_{i=1}^n \frac{\partial(y^i \circ \phi^{-1})}{\partial x_k} \Big|_p \frac{\partial}{\partial y^i} \Big|_p$$

on $U \cap V$.

4.2 Equivalent Characterizations of the Tangent Space

Lemma 4.2.1

Let M be a smooth manifold. Then

$$m_p = \{f \in C_{M,p}^\infty \mid f(p) = 0\}$$

is a maximal ideal of $C_{M,p}^\infty$.

Proposition 4.2.2

Let M be a smooth manifold. Then the following are true.

- There is a natural $\mathbb{R}$ -linear isomorphism

$$T_p(M) \cong \text{Der}_{\mathbb{R}}(C_{M,p}^\infty, \mathbb{R})$$

given by $[\gamma] \mapsto X_{\gamma,p}$ where $\mathbb{R}$ is a $C_{M,p}^\infty$ -module via the isomorphism $\mathbb{R} \cong \frac{C_{M,p}^\infty}{m_p}$.

- There is a natural $\mathbb{R}$ -linear isomorphism

$$\text{Der}_{\mathbb{R}}(C_{M,p}^\infty, \mathbb{R}) \cong \left(\frac{m_p}{m_p^2} \right)^*$$

given by restriction of a derivation d to the ideal m_p and then taking quotient.

4.3 Differential of Smooth Maps

Definition 4.3.1 (Differential of a Map)

Let M, N be smooth manifolds. Let $f : M \rightarrow N$ be a smooth map. Let $p \in M$. Define the differential of f at p to be the map $(f_*)_p : T_p(M) \rightarrow T_p(N)$ given by

$$(f_*)_p(X_p) = X_p(- \circ f)$$

Theorem 4.3.2

Let M, N be smooth manifolds. Let $f : M \rightarrow N$ be a smooth map. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart of M containing p . Let $(V, y^1, \dots, y^m)$ be a chart of N containing $f(p)$. Then $(f_*)_p$ is given by the matrix that is the Jacobian:

$$(f_*)_p = \begin{pmatrix} \left. \frac{\partial}{\partial x^1} \right|_p (y^1 \circ f) & \cdots & \left. \frac{\partial}{\partial x^n} \right|_p (y^1 \circ f) \\ \vdots & \ddots & \vdots \\ \left. \frac{\partial}{\partial x^1} \right|_p (y^m \circ f) & \cdots & \left. \frac{\partial}{\partial x^n} \right|_p (y^m \circ f) \end{pmatrix}$$

To compute the differential of a map applied to a tangent vector, write $X_{\gamma,p}$ in terms of the basis at (U, ϕ) . For example, $X_{\gamma,p}$ can be given by $(a_1(p) \cdots a_n(p))^T$. Then apply the linear transformation $(f_*)_p$ to receive an $m \times 1$ vector:

$$(f_*)_p(X_{\gamma,p}) = \begin{pmatrix} \left. \frac{\partial(y_1 \circ f \circ \phi^{-1})}{\partial x_1} \right|_{\phi(p)} & \cdots & \left. \frac{\partial(y_1 \circ f \circ \phi^{-1})}{\partial x_n} \right|_{\phi(p)} \\ \vdots & \ddots & \vdots \\ \left. \frac{\partial(y_m \circ f \circ \phi^{-1})}{\partial x_1} \right|_{\phi(p)} & \cdots & \left. \frac{\partial(y_m \circ f \circ \phi^{-1})}{\partial x_n} \right|_{\phi(p)} \end{pmatrix} \begin{pmatrix} a_1(p) \\ \vdots \\ a_n(p) \end{pmatrix}$$

The resulting vector is then the tangent vector of $T_{f(p)}(N)$ in terms of the basis elements around the chart of N .

Proposition 4.3.3 (Chain Rule) Let $f : M \rightarrow N$ and $g : N \rightarrow P$ be a smooth map on manifolds M, N, P . Let $p \in M$. Then

$$((g \circ f)_*)_p = (g_*)_{f(p)} \circ (f_*)_p$$

where the differential is taken as the tangent space over $p, f(p)$ and $g(f(p))$.

The following corollary will use the fact that the differential of the identity map is again the identity. This can be seen using the Jacobian description of the differential.

Corollary 4.3.4 Let $f : M \rightarrow N$ be a diffeomorphism of smooth manifolds M and N . Then for any $p \in M$, $(f_*)_p : T_p(M) \rightarrow T_{f(p)}(N)$ is an isomorphism of vector spaces.

Proof If $f : M \rightarrow N$ is a diffeomorphism, then f has a smooth inverse f^{-1} . Then

$$\text{id}_{T_p(M)} = (\text{id}_*)_p = ((f^{-1} \circ f)_*)_p = ((f^{-1})_*)_{f(p)} \circ (f_*)_p$$

and similarly for $\text{id}_{T_q(N)} = (f_*)_{f^{-1}(q)} \circ ((f^{-1})_*)_q$ show that $(f_*)_p$ is an isomorphism. ■

The chain rule together with the fact that $(\text{id}_*)_p = \text{id}_{T_p(M)}$ show that the map $f \mapsto (f_*)_p$ is functorial for any choice of $p \in M$.

Using the differential, we can give an alternate proof for the basis of the vector space $T_p(M)$.

For each $\phi : U \rightarrow \mathbb{R}^n$ a chart on M , ϕ is a diffeomorphism onto its image so that the differential $\phi_* :$

$T_p(M) \rightarrow T_{\phi(p)}(\mathbb{R}^n)$ is an isomorphism. In particular, the tangent vector $\frac{\partial}{\partial x^k} \Big|_p$ is sent to

$$\phi_* \left(\frac{\partial}{\partial x^k} \Big|_p \right) (g) = \frac{\partial(g \circ \text{id}_{\phi(U)})}{\partial x_k} \Big|_{\phi(p)} = \frac{\partial g}{\partial x_k} \Big|_{\phi(p)}$$

for any choice $g \in C_{\mathbb{R}^n, \phi(p)}^\infty$. Notice that the differential changed operator because x^k refers to coordinates on the manifold while x_k refers to coordinates on $\mathbb{R}^n$. Now the isomorphism ϕ_* carries basis to basis and in multivariable calculus we have already seen that elements of the latter form a basis for $T_p(\mathbb{R}^n)$.

4.4 The Preimage Theorem

Definition 4.4.1 (Regular Values) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds. We say that $q \in N$ is a regular value if for $p \in f^{-1}(q)$, $(f_*)_p$ is surjective. Otherwise it is said to be a critical value.

Definition 4.4.2 (Regular Level Set) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds. We say that a level set $f^{-1}(q)$ for $q \in N$ is regular if for all $p \in f^{-1}(q)$, p is a regular value.

The preimage theorem is a variation of the inverse function theorem, except that it is applied to a manifold instead of $\mathbb{R}^n$.

Theorem 4.4.3 (The Preimage Theorem) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. Let $q \in N$. If $f^{-1}(q)$ is a non empty regular level set, then $f^{-1}(q)$ is an embedded submanifold of N of dimension $m - n$.

Note the similarities of the preimage theorem and proposition 2.2.3. Proposition 2.2.3 is actually a special case of the preimage theorem. By choosing $N = \mathbb{R}^c$, we obtain proposition 2.2.3. In particular, the condition on the Jacobian being non-zero is the same as requiring that $(f_*)_p$ to be surjective.

Proposition 4.4.4 Let $f : M \rightarrow N$ be a smooth map of manifolds such that $V = f^{-1}(c)$ is a regular level set. Then

$$T_p(V) = \ker((f_*)_p)$$

for any $p \in V$.

Proof Let $X_{\gamma,p} \in T_p(V)$. Then we have that

$$\begin{aligned} (f_*)_p(X_{\gamma,p})(g) &= X_{\gamma,p}(g \circ f) \\ &= \frac{d(g \circ f \circ \gamma)}{dt} \Big|_{t=0} \\ &= \frac{dg(q)}{dt} \Big|_{t=0} \\ &= 0 \end{aligned}$$

for any choice of $g \in C_{N, f(p)}^\infty$ so that $T_p(V) \subseteq \ker((f_*)_p)$. Since c is a regular value, $(f_*)_p$ is surjective and so the rank of the linear map is $n = \dim(N)$. By rank nullity theorem, the kernel $\ker((f_*)_p)$ has dimension $m - n$. The preimage theorem also tells us that $T_p(V)$ has dimension $m - n$ and so we are done. ■

Theorem 4.4.5 (Sard's Theorem) Let M and N be smooth manifolds. Let $f : M \rightarrow N$ be a smooth map. Then the set of critical values of f has measure 0 in N .

4.5 Rank of the Differential

The rank of the differential gives crucial information to how smooth maps of manifolds give rise to submanifolds. Consider a smooth map $f : M \rightarrow N$ between smooth manifolds. Let (U, ϕ) and $(V, \psi = (y^1, \dots, y^n))$ be charts on p and $f(p)$ respectively. We have seen that the map $(f_*)_p$ is given by the Jacobian $\left(\frac{\partial(y^i \circ f \circ \phi^{-1})}{\partial x^j} \Big|_p \right)_{n \times m}$. It is easy to see that

$$\begin{aligned} (f_*)_p \text{ is injective} &\iff m \leq n \text{ and } \text{rank} \left(\left(\frac{\partial(y^i \circ f \circ \phi^{-1})}{\partial x^j} \Big|_{\phi(p)} \right)_{n \times m} \right) = m \\ (f_*)_p \text{ is surjective} &\iff m \geq n \text{ and } \text{rank} \left(\left(\frac{\partial(y^i \circ f \circ \phi^{-1})}{\partial x^j} \Big|_{\phi(p)} \right)_{n \times m} \right) = n \end{aligned}$$

Theorem 4.5.1 (Immersion Theorem) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is injective, then there are charts (U, ϕ) of M and (V, ψ) of N such that

$$(\psi \circ f \circ \phi^{-1})(x_1, \dots, x_m) = (x_1, \dots, x_m, 0, \dots, 0)$$

In other words, locally at each point, f is an inclusion $\mathbb{R}^m \rightarrow \mathbb{R}^n$.

Proof Suppose that $(f_*)_p$ is injective. Let (U, ϕ) be a chart on p and $(V, \psi = y_1, \dots, y_n)$ a chart on $f(p)$. Using proposition 3.2.2 gives us the Jacobian of $\psi \circ f \circ \phi^{-1}$. Since $(f_*)_p$ is injective, we have that the matrix has rank m . This means that we can choose indices $i_1, \dots, i_m$ from $\{1, \dots, n\}$ so that the submatrix with rows $i_1, \dots, i_m$, calling it A , is invertible. By the inverse function theorem, A gives a local diffeomorphism. Notice that functions $(y_{i_1} \circ f, \dots, y_{i_m} \circ f)$ is a function from U to $\mathbb{R}^m$. Since it is also diffeomorphic, it is actually a chart on M . This completes the proof. ■

Theorem 4.5.2 (Submersion Theorem) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is surjective, then there are charts (U, ϕ) of M and (V, ψ) of N such that

$$(\psi \circ f \circ \phi^{-1})(x_1, \dots, x_n, x_{n+1}, \dots, x_m) = (x_1, \dots, x_n)$$

In other words, locally at each point, f is a projection $\mathbb{R}^m \rightarrow \mathbb{R}^n$.

Proof Suppose that $(f_*)_p$ is surjective. Let (U, ϕ) and (V, ψ) be charts around $p \in M$ and $f(p) \in N$ respectively. Then we know that f_* is given by the Jacobian matrix of $\psi \circ f \circ \phi^{-1}$. Since $(f_*)_p$ is surjective the $n \times m$ Jacobian matrix has rank n . This means that we can choose n columns from the $n \times m$ matrix for which the submatrix A of these n columns are invertible. By the inverse function theorem, A gives a local diffeomorphism. ■

Corollary 4.5.3 Let $f : M \rightarrow N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is bijective, then there are charts (U, ϕ) and (V, ψ) of p and $f(p)$ respectively such that $\psi \circ f \circ \phi^{-1} : \phi(U) \subseteq \mathbb{R}^m \rightarrow \psi(V) \subseteq \mathbb{R}^n$ is a homeomorphism. In particular, this means that $m = n$.

Proof Combine the immersion theorem and submersion theorem for the result. ■

We give names to the case when $(f_*)_p$ is injective or surjective for all points p on the manifold. In particular, this means that the immersion / submersion theorem applies to all points on the manifold.

Definition 4.5.4 (Immersion and Submersions) Let $f : M \rightarrow N$ be a smooth map of manifolds.

- If $(f_*)_p$ is injective for all $p \in M$, then f is called an immersion. We then say that M is immersed in N by f .
- If $(f_*)_p$ is surjective for all $p \in M$, then f is called a submersion.

Note that the definition of submersions and regular values are slightly different. Namely regular values are defined for points in the codomain while the condition on submersion checks every point on the domain of the smooth map.

Definition 4.5.5 (Embeddings) Let $f : M \rightarrow N$ be a smooth map of manifolds. We say that f is an embedding of M into N if f is injective, is an immersion and is homeomorphic to its image $f(M)$.

Proposition 4.5.6 Let M be a smooth manifold. Let N be a subset of M . Then N is an embedded submanifold of M if and only if the inclusion map $\iota : N \rightarrow M$ is an embedding.

Note that the condition on homeomorphism is necessary for $f(M)$ to be an embedded submanifold of N in the definition of embeddings only when M is not compact. This can be seen with the example of embedding a line to a torus. Consider

$$\varphi : \mathbb{R} \rightarrow L \subseteq \mathbb{R}^2 \rightarrow \frac{\mathbb{R}^2}{\mathbb{Z}^2} \cong T^2$$

where L is a line passing through the origin of $\mathbb{R}^2$ with irrational slope. Then $\varphi(\mathbb{R})$ is dense in the torus T^2 .

5 Some Examples of Smooth Manifolds

5.1 Manifolds in the Set of Real Matrices

Proposition 5.1.1 Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are smooth submanifolds of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$.

- The general linear group $GL(n, \mathbb{R})$
- The orthogonal group $O(n, \mathbb{R})$
- The special linear group $SL(n, \mathbb{R})$
- The special orthogonal group $SO(n, \mathbb{R})$.

Proof

- Notice that the determinant map $M_n(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $A \mapsto \det(A)$ is smooth. Indeed the determinant is a degree n polynomial in each entry of an $n \times n$ matrix and polynomials $\mathbb{R}^2 \rightarrow \mathbb{R}$ are smooth. Then $GL(n, \mathbb{R})$ is precisely the set $\det^{-1}(\mathbb{R} \setminus \{0\})$ so that $GL(n, \mathbb{R})$ is an open subset of $M_n(\mathbb{R})$. By prp3.1.3, $GL(n, \mathbb{R})$ is a smooth manifold of dimension n^2 . ■

Proposition 5.1.2 Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are embedded submanifolds of $GL(n, \mathbb{R})$.

- The orthogonal group $O(n, \mathbb{R})$, with dimension $n^2 - (n^2 + n)/2 = (n^2 - n)/2$.
- The special linear group $SL(n, \mathbb{R})$
- The special orthogonal group $SO(n, \mathbb{R})$

Proof We first consider $O(n, \mathbb{R})$. Let S_n denote the set of symmetric matrices in $M_n(\mathbb{R})$. Since S is a finite dimensional vector space over $\mathbb{R}$, it is also a smooth manifold of the same dimension. Let $F : M_n(\mathbb{R}) \rightarrow S_n$ be the map $M \mapsto M^T M$. Notice that $O(n, \mathbb{R}) = F^{-1}(I_n)$. Let $A \in F^{-1}(I_n) = O(n)$. We can calculate the differential $(dF_*)_A : \mathbb{R}^{n^2} \cong M_n(\mathbb{R}) \rightarrow S_n \cong \mathbb{R}^{(n^2+n)/2}$ (vector space isomorphisms) by

$$(dF_*)_A(X) = \left. \frac{d}{dt} F(A + tX) \right|_{t=0} = \left. \frac{d}{dt} (A + tX)^T (A + tX) \right|_{t=0} = A^T X + X^T A$$

for any $X \in M_n(\mathbb{R})$. I claim that this linear map is surjective. Let $B \in S_n$ be a symmetric matrix. Using $X = \frac{1}{2}AB$, we have that

$$(dF_*)_A(AB) = A^T \left(\frac{1}{2}AB \right) + \left(\frac{1}{2}AB \right)^T A = \frac{1}{2}(B + B^T) = B$$

because A is orthogonal and B is symmetric. The preimage theorem is satisfied, hence $O(n, \mathbb{R})$ is an embedded submanifold of $GL(n, \mathbb{R})$ of dimension $\dim(O(n, \mathbb{R})) = n^2 - (n^2 + n)/2 = (n^2 - n)/2$. ■

6 Vector Bundles on Smooth Manifolds

While the previous section introduced general vector bundles over an arbitrary topological space, the concept of smooth bundles are exclusive solely to smooth manifolds. Indeed the notion of smoothness exists only in smooth manifolds when we say that the transition maps of the charts are smooth functions from Euclidean spaces.

6.1 Smooth Vector Bundles

Definition 6.1.1 (Smooth Vector Bundles) Let $p : E \rightarrow M$ be an $\mathbb{R}$ -vector bundle of rank r . The vector bundle is said to be smooth if E, M are smooth manifolds, p is a smooth map of manifolds and the local trivializations (U, ϕ) are diffeomorphisms from $p^{-1}(U)$ to $U \times \mathbb{R}^r$.

Diffeomorphisms here means the following. Let $(\tilde{U}, \tilde{\phi})$ be a chart on E and (U, ϕ) a chart on M . Then if (V, ψ) are local trivializations, we require that

$$\phi \circ \psi \circ \tilde{\phi}^{-1} : \tilde{\phi}(\tilde{U}) \subseteq \mathbb{R}^e \rightarrow \phi(U) \subseteq \mathbb{R}^m$$

to be diffeomorphisms whenever $\psi(V) \subseteq U$.

Proposition 6.1.2 Let $p : E \rightarrow M$ be a smooth vector bundle. Then the set of all C^∞ sections

$$\Gamma(E) = \{s : M \rightarrow E \mid s \text{ is a smooth section}\}$$

is a vector space over $\mathbb{R}$ and a module over the ring $C^\infty(M)$.

Proof It is clear that $1 + s_2$ and fs is a section. It remains to check that they are smooth sections. Let $p \in M$. Let (U, ϕ) be a smooth local trivialization of $p : E \rightarrow M$ containing p . Suppose that $\phi \circ s : M \rightarrow U \times \mathbb{R}^k$ is given by

$$(\phi \circ s_1)(q) = (q, a_1(q), \dots, a_k(q))$$

and similarly $(\phi \circ s_2)(q) = (q, b_1(q), \dots, b_k(q))$ for $q \in U$. Since s_1, s_2 are smooth maps, a_i and b_j are smooth functions on U . Since ϕ is linear on each fiber,

$$(\phi \circ (s_1 + s_2))(q) = (q, (a_1 + b_1)(q), \dots, (a_k + b_k)(q))$$

so that $s_1 + s_2$ is a smooth map on U and hence at p . Since p is an arbitrary point we conclude.

The proof for the module structure is similar to the above. ■

Definition 6.1.3 (Smooth Frames) Let $\pi : E \rightarrow M$ be a smooth vector bundle. A frame for p is a collection of local sections $s_1, \dots, s_r : U \rightarrow E$ over $U \subseteq M$ such that for each $x \in M$, the set

$$\{s_1(x), \dots, s_r(x)\}$$

is a basis for the fiber $E_x = p^{-1}(x)$. A frame is said to be smooth if $s_1, \dots, s_r$ are C^∞ sections of E over U .

We write $\bar{e}_1, \dots, \bar{e}_r : M \rightarrow M \times \mathbb{R}^r$ for the trivial bundle $\pi : M \times \mathbb{R}^r \rightarrow M$, where each $\bar{e}_i : M \rightarrow M \times \mathbb{R}^r$ are defined by

$$\bar{e}_i(p) = (p, e_i)$$

for e_i the standard i th basis element of $\mathbb{R}^r$. In particular, for a local trivialization $\phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$ of a general vector bundle, we can define a smooth frame over the local trivialization in terms of frame of the trivial bundle. This is the frame $t_1, \dots, t_r : U \rightarrow E$ defined by $t_i(p) = \phi^{-1}(\bar{e}_i(p)) = \phi^{-1}(p, e_i)$ for each i .

Lemma 6.1.4 Let $\pi : E \rightarrow M$ be a smooth vector bundle. Let $\phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$ be a local trivialization of an open set $U \subseteq M$. Let $t_1, \dots, t_r : U \rightarrow E$ be the smooth frame over the trivialization. Then a local section

$$s = \sum_{k=1}^r b_k t_k$$

of E over U is smooth if and only if its coefficients b_k relative to the frame $t_1, \dots, t_r$ are smooth.

Proof If the coefficients are smooth functions then by proposition 5.1.2, s is smooth. Conversely, suppose that the section $s = \sum_{k=1}^r b_k t_k$ of E over U is smooth. Then $\phi \circ s$ is smooth. Notice that

$$(\phi \circ s)(p) = \sum_{k=1}^r b_k(p) \phi(t_k(p)) = \sum_{k=1}^r b_k(p)(p, e_k) = \left(p, \sum_{k=1}^r b_k(p) e_k \right)$$

Thus $b_k(p)$ for each k are simply the fiber coordinates of $s(p)$ relative to the trivialization ϕ . Since $\phi \circ s$ is smooth, all the b_k are smooth. ■

Proposition 6.1.5 Let $\pi : E \rightarrow M$ be a smooth vector bundle. Let $U \subseteq M$ be an open subset of M . Suppose that $s_1, \dots, s_r$ is a smooth frame for E over U . Then a section

$$s = \sum_{k=1}^r c^k s_k$$

of E over U is smooth if and only if the coefficients $c_k : U \rightarrow \mathbb{R}$ are smooth functions.

Proof If each c_k are smooth functions, then by proposition 5.1.2, s is smooth. Conversely, suppose that $s = \sum_{k=1}^r c^k s_k$ is a smooth section. Let $p \in U$ and choose a trivializing open set $V \subset U$ for E containing p with smooth trivialization $\phi : \pi^{-1}(V) \rightarrow V \times \mathbb{R}^r$. Let $t_1, \dots, t_r$ be the smooth frame of ϕ . This means that $t_i(p) = \phi^{-1}(\bar{e}_i(p)) = \phi^{-1}(p, e_i)$. Then we have that

$$\sum_{k=1}^r b_k t_k = s = \sum_{k=1}^r \sum_{j=1}^r c_j a_j^k t_k$$

Comparing coefficients give $b_k = \sum_{j=1}^r c_j a_j^k$. In matrix notation, we have that

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_r \end{pmatrix} = A \begin{pmatrix} c_1 \\ \vdots \\ c_r \end{pmatrix} = Ac$$

At each point of V , A is the transition matrix between the two bases and so is invertible. By Cramer's rule, A^{-1} is a matrix of smooth functions on V . Hence $c = A^{-1}b$ is a column vector of smooth functions on V . This proves that $c_1, \dots, c_r$ are smooth functions at $p \in U$. Since p is arbitrary, the coefficients c_j are smooth functions on U . ■

6.2 The Tangent Bundle

Definition 6.2.1 (The Tangent Bundle) Let M be a smooth manifold. Define the tangent bundle of M to be the disjoint union

$$TM = \coprod_{p \in M} T_p M$$

Define the projection map $\pi : TM \rightarrow M$ by $\pi(v) = p$ if $v \in T_p M$.

Definition 6.2.2 (Local Trivialization Maps) Let M be a smooth manifold of dimension n . Let

(U, ϕ) be a chart on M . Let

$$TU = \coprod_{q \in U} T_q(M)$$

Let $v \in TU$ and let $p = \pi(v)$. Then $v \in T_p M$ is given locally on a chart by

$$v = \sum_{k=1}^n c_k \frac{\partial}{\partial x^k} \Big|_p$$

for smooth functions $c_k : U \rightarrow \mathbb{R}$. Define the local trivialization map

$$\tilde{\phi} : TU \rightarrow U \times \mathbb{R}^n$$

by the formula

$$\tilde{\phi}(v) = (p, c_1(p), \dots, c_n(p))$$

Lemma 6.2.3 Let M be a smooth manifold of dimension n . Let (U, ϕ) be a chart on M . Then the local trivialization maps

$$\tilde{\phi} : TU \rightarrow U \times \mathbb{R}^n$$

is a bijection.

Proof It is easy to see that the map defined by

$$(\mathbf{x}, c_1, \dots, c_n) \mapsto \left(\phi^{-1}(\mathbf{x}), \sum_{k=1}^n c_k \frac{\partial}{\partial x^k} \Big|_p \right)$$

is an inverse to $\tilde{\phi}$. ■

Theorem 6.2.4 (Topology on the Tangent Bundle) Let M be a smooth manifold of dimension n . Then TM is a topological space with basis

$$\mathcal{B} = \bigcup_{\alpha} \left\{ A \subseteq TM \mid A \text{ is open in } T(U_{\alpha}) \text{ and } U_{\alpha} \text{ is a coordinate open set in } M \right\}$$

where we say that A is open in $T(U_{\alpha})$ if and only if $\tilde{\phi}(A)$ is open in $\phi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$.

Proof Firstly, notice that if $V \subseteq U$ is open, then $\phi(V) \times \mathbb{R}^n$ is an open subset of $\phi(U) \times \mathbb{R}^n$ so that the relative topology on $T(V)$ as a subset of TU is the same as the topology induced from the bijection $\tilde{\phi}|_{T(V)} : T(V) \rightarrow \phi(V) \times \mathbb{R}^n$.

Let $\{(U_{\alpha}, \phi_{\alpha})\}$ be the maximal atlas for M . Then

$$TM = \bigcup_{\alpha} T(U_{\alpha}) \subseteq \bigcup_{A \in \mathcal{B}} A \subseteq TM$$

so that equality holds everywhere.

Let A be open in $T(U)$ and B open in $T(V)$. The goal is to show that $A \cap B$ is open in $T(U \cap V)$. Since $T(U \cap V)$ is a subspace of $T(U)$, by definition of relative topology, $A \cap T(U \cap V)$ is open in $T(U \cap V)$. Similarly, $B \cap T(U \cap V)$ is open in $T(U \cap V)$ so that

$$A \cap B \subseteq T(U) \cap T(V) = T(U \cap V)$$

is open in $T(U \cap V)$. ■

Proposition 6.2.5 Let M be a smooth manifold of dimension n . Then the tangent bundle TM is a vector bundle of dimension n .

Proof

■

Theorem 6.2.6 Let M be a manifold. Then the tangent space TM is a manifold of dimension $2n$.

Proof Let $\{(U_\alpha, \phi_\alpha)\}$ be the maximal atlas for M and $\mathcal{B}$ a countable basis for M . For each coordinate open set U_α and $p \in U_\alpha$, choose a basic open set $B_{p,\alpha} \in \mathcal{B}$ such that $p \in B_{p,\alpha} \subset U_\alpha$. The collection $\{B_{p,\alpha}\}$ without duplicate elements is a subcollection of $\mathcal{B}$ and is countable. For any open set U in M and $p \in U$, there is a coordinate open set U_α such that $p \in U_\alpha \subset U$. Hence $p \in B_{p,\alpha} \subset U$ so that $\{B_{p,\alpha}\}$ is a countable basis for M . Let $\{U_i \mid i \in \mathbb{N}\}$ be a countable basis of M consisting of coordinate open sets. Let ϕ_i be the coordinate map on U_i . Since $T(U_i)$ is homeomorphic to the open subset $\phi_i(U_i) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$ and any subset of a Euclidean space is second countable, $T(U_i)$ is second countable. For each i , choose a countable basis $\{B_{i,j} \mid j \in \mathbb{N}\}$ for $T(U_i)$. Then $\{B_{i,j} \mid i, j \in \mathbb{N}\}$ is a countable basis for $T(M)$ so that (TM) is second countable.

For Hausdorffness,

Finally, if $\{(U_\alpha, \phi_\alpha)\}$ is a smooth atlas on M , then $\{(T(U_\alpha), \widetilde{\phi}_\alpha)\}$ is a smooth atlas for $T(M)$. It is clear that $T(M) = \bigcup_\alpha T(U_\alpha)$. It remains to check that they are smoothly compatible. Recall that by proposition 3.1.7, if $(U_\alpha, \phi_\alpha = (x^1, \dots, x^n))$ and $(U_\beta, \phi_\beta = (y^1, \dots, y^n))$ are charts on M and $p \in U \cap V$ such that

$$X_p = \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k} \Big|_p = \sum_{k=1}^n b_k(p) \frac{\partial}{\partial y^k} \Big|_p$$

are the two representations of the same tangent vector in the two charts, then we have

$$a_i(p) = \sum_{j=1}^n b_j(p) \frac{\partial x^i}{\partial y^j} \Big|_p \quad \text{and} \quad b_i(p) = \sum_{k=1}^n a_k(p) \frac{\partial y^i}{\partial x^k} \Big|_p$$

Now we have

$$\widetilde{\phi}_\beta \circ \widetilde{\phi}_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n \rightarrow \phi_\beta(U_\alpha \cap U_\beta) \times \mathbb{R}^n$$

is given by

$$(\phi_\alpha(p), a_1(p), \dots, a_n(p)) \mapsto \left(p, \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k} \Big|_p \right) \mapsto (\phi_\beta(p), b_1(p), \dots, b_n(p))$$

where each $b_i(p)$ is given by the formula

$$b_i(p) = \sum_{k=1}^n a_k(p) \frac{\partial y^i}{\partial x^k} \Big|_p = \sum_{k=1}^n a_k(p) \frac{\partial (\phi_\beta \circ \phi_\alpha^{-1})^i}{\partial x_j} \Big|_{\phi_\alpha(p)}$$

where $(\phi_\beta \circ \phi_\alpha^{-1})^i$ means the i th component of the map $\phi_\beta \circ \phi_\alpha^{-1}$. Therefore $\widetilde{\phi}_\beta \circ \widetilde{\phi}_\alpha^{-1}$ is smooth. Thus we conclude. ■

6.3 Vector Fields

Definition 6.3.1 (Vector Fields) Let M be a manifold and $T_p(M)$ the tangent space of p in M . A vector field $X : M \rightarrow T_p(M)$ assigns for each $p \in M$ an element in the tangent space. This means that X is a section of the tangent bundle $\pi : T(M) \rightarrow M$.

Definition 6.3.2 (Smooth Vector Fields) Let M be a smooth manifold. A vector field X is smooth if it is smooth as a map from M to TM . Denote the set of all smooth vector fields on M by $\mathfrak{X}(M)$.

From now on, we will only consider smooth vector fields as they carry the most structure for our smooth manifolds. $\mathfrak{X}(M)$ has a rich structure in terms of algebra.

Proposition 6.3.3 Let M be a smooth manifold. Then $\mathfrak{X}(M)$ is an $\mathbb{R}$ -vector space and a $C^\infty(M)$ -module.

In particular, $\mathfrak{X}(M)$ is also a Lie algebra.

Under local coordinates, we can give an explicit formula for a vector field.

Proposition 6.3.4 Let M be a manifold and $(U, \phi = (x_1, \dots, x_n))$ be a chart on M . A vector field

$$X = \sum_{k=1}^n a_k \frac{\partial}{\partial x_k}$$

on U is smooth if and only if the coefficient functions $a_k : U \rightarrow \mathbb{R}$ are smooth on U .

Proof By proposition 5.1.5. ■

6.4 Integral Curves of Vector Fields

Definition 6.4.1 (Integral Curves) Let M be a manifold and X a smooth vector field over M . A curve $\gamma(t)$ in M is said to be an integral curve of X if for every parameter value t_0 , the vector $X_{\gamma(t_0)}$ is tangent to the curve $\gamma(t)$ at $\gamma(t_0)$.

Proposition 6.4.2 Let X be a vector field on a manifold M . For any $p \in M$, there exists a unique integral curve $x(t)$ of X defined for $|t| < \epsilon$ for some $\epsilon > 0$ such that $p_0 = x(0)$.

Definition 6.4.3 (1-parameter Group of Diffeomorphisms) Let M be a manifold. A 1-parameter group of diffeomorphisms (global flow) is a map $\Psi : \mathbb{R} \times M \rightarrow M$ with $\Psi(t, p) = \Psi_t(p) \in M$ such that the following conditions are satisfied.

- For all $t \in \mathbb{R}$, $\Psi_t : M \rightarrow M$ is a diffeomorphism of M
- For all $t, s \in \mathbb{R}$, $\Psi_{t+s} = \Psi_t \circ \Psi_s$

Notice that every 1-parameter group of diffeomorphisms naturally induce a vector field over M .

Definition 6.4.4 (Local 1-parameter Group of local Diffeomorphisms) Let M be a manifold and $U \subseteq M$ be open. A local 1-parameter group of local diffeomorphisms (global flow) is a map $\Psi : (-\epsilon, \epsilon) \times U \rightarrow M$ with $\Psi(t, p) = \Psi_t(p) \in M$ such that the following conditions are satisfied.

- For all $t \in (-\epsilon, \epsilon)$, $\Psi_t : U \rightarrow \Psi_t(U)$ is a diffeomorphism
- For all $t, s, t + s \in (-\epsilon, \epsilon)$, $\Psi_{t+s} = \Psi_t \circ \Psi_s$

Similar to the global version, this local version naturally induces a vector field on U .

Proposition 6.4.5 Let X be a smooth vector field over a manifold M . For each point $p_0 \in M$, there is an open neighbourhood U of p_0 and $\epsilon > 0$ and a local 1-parameter group of local diffeomorphisms

$$\Psi_t : U \rightarrow M$$

inducing X on U .

Moreover, if two local 1-parameter group of local diffeomorphisms induce the same vector field over U , then they coincide.

Proof Let $(U, \phi = (x_1, \dots, x_n))$ be a chart on M centered at p . Write

$$X = \sum_{i=1}^n \xi_i(x_1, \dots, x_n) \frac{\partial}{\partial x_i}$$

locally on U . Consider the system of ODEs

$$\frac{df_i}{dt} = \xi_i(f_1(t), \dots, f_n(t))$$

for $1 \leq i \leq n$ and unknown functions $f_1, \dots, f_n$. By the fundamental theorem for system of ODEs, there are unique functions $f_1(t, x), \dots, f_n(t, x)$ defined for all $x = (x_1, \dots, x_n)$ with $|x_i| < \delta_1$ and $|t| < \varepsilon_1$ for δ_1, ε_1 some small positive real numbers, and depending smoothly on t and x , which form a system of solutions of the differential equations for each fixed x and satisfy the initial conditions $f_i(0, x) = x_i$.

Set $\Phi(t, x) = (f_1(t, x), \dots, f_n(t, x))$ for $|t| < \varepsilon_1$ and $x \in U_1 = \{x \in M \mid |x_i| < \delta_i \text{ for all } i\}$. If $|t|, |s|$ and $|t + s|$ are all less than ε_1 and both x and $\Phi(s, x) = \Phi_s(x)$ are in U_1 , then the functions $g_i(t) = f_i(t + s, x)$ are a solution to the initial system of differential equations with initial conditions $g_i(0) = f_i(s, x)$. Thus by the uniqueness of solutions, we have

$$g_i(t) = f_i(t, \Phi(s, x))$$

Thus $\Phi_t(\Phi_s(x)) = \Phi_{t+s}(x)$. Moreover, there exists $\delta > 0$ and $\varepsilon > 0$ such that putting $U = \{x \in M \mid |x_i| < \delta \text{ for all } i\}$, we have $\Phi_t(U) \subset U_1$ for $|t| < \varepsilon$: Indeed, this follows because Φ_0 is the identity on U_1 . Therefore, we have that

$$\Phi_t(\Phi_{-t}(x)) = \Phi_0(x) = x$$

for all $x \in U$ and $|t| < \varepsilon$ and all Φ_t are diffeomorphisms of U onto $\Phi_t(U)$. Thus Φ is a local 1-parameter group of local diffeomorphisms defined on $I_\varepsilon \times U$ inducing X . ■

Definition 6.4.6 (Complete Vector Fields) We say that a vector field X over a manifold M is complete if X is induced by a 1-parameter group of diffeomorphisms (global flow).

Proposition 6.4.7 Every vector field X on a compact manifold M is complete.

Proof For every point p , let $U(p)$ be a neighbourhood of p and $\varepsilon(p) > 0$ such that there exists a local 1-parameter group of local diffeomorphisms Φ defined on $I_{\varepsilon(p)} \times U(p)$ inducing X . Since M is compact, finitely many of the $U(p)$, say $U(p_1), \dots, U(p_k)$ cover M . Set $\varepsilon = \min\{\varepsilon(p_1), \dots, \varepsilon(p_k)\}$. We see that $\Phi_t(p)$ can be defined on $I_\varepsilon \times M$, hence on all of $\mathbb{R} \times M$. ■

7 Differential Forms and its Applications

In considering a smooth section on the tangent bundle we obtain the notion of smooth vector fields. The smooth section on the cotangent bundle will lead to smooth 1-forms. However, the theory of k -forms also provide a notion of integration on manifolds. In particular, Stoke's theorem on n -forms generalizes all notions of divergence and curl, including Green's theorem and the divergence theorem. By considering all differential k -forms as a whole, they also contain topological information including the number of holes of space. This odd similarity with that of cohomology in Algebraic Topology leads to de Rham's theorem, which states that there is an isomorphism between singular cohomology of a smooth manifold and de Rham cohomology.

7.1 Differential 1-Forms

Definition 7.1.1 (Differential 1-Forms) Let M be a smooth manifold. A differential 1-form on M is a section $s : M \rightarrow T^*(M)$ of the cotangent bundle $p : T^*(M) \rightarrow M$. A smooth differential 1-form on M is a smooth section of the cotangent bundle.

To write it in local coordinates, we make use of also the notion of cotangent spaces.

Definition 7.1.2 (Cotangent Space) Let M be a smooth manifold. Let $p \in M$. Define the cotangent space of M to be

$$T_p^*(M) = \text{Hom}_{\mathbb{R}}(T_p(M), \mathbb{R})$$

An element of the cotangent space is called a cotangent vector. The collection of all cotangent spaces for each point is in particular, the cotangent bundle.

Definition 7.1.3 (Differential of a Function) Let M be a smooth manifold. For each $f \in C^\infty(M)$, define $df : M \rightarrow T^*(M)$ to be the 1-form such that for each $p \in M$, $(df)_p$ is a map from $T_p(M)$ to $\mathbb{R}$ where

$$(df)_p(X_p) = X_p(f)$$

Note that the above definition makes sense. Firstly, df is a map from M so to each point p on M , we assign a covector $(df)_p$. But a covector is none other than an $\mathbb{R}$ -linear map from $T_p(M)$ to $\mathbb{R}$ so we need to define what $(df)_p$ does to each $X_p \in T_p(M)$.

Recall from linear algebra that V and its dual space V^* has the same dimension if V is finite dimensional. This applies also to the case $T_p^*(M)$. Moreover, we can compute a basis for $T_p^*(M)$ in terms of differential of some particular functions in $C^\infty(M)$, at least locally.

Proposition 7.1.4 Let M be a smooth manifold. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart at $p \in M$. Write dx^k the differential of the smooth function $x^k : U \rightarrow \mathbb{R}$ (so that it is a 1-form). Then

$$\{(dx^1)_p, \dots, (dx^n)_p\}$$

is the dual basis in $T_p^*(M)$ of $\left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p \right\}$.

Proof By definition, we have that

$$(dx^j)_p \left(\left. \frac{\partial}{\partial x^i} \right|_p \right) = \left. \frac{\partial x^j}{\partial x^i} \right|_p = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and so we conclude. ■

Using the local basis of the cotangent space, we can also express the differential of a function and smooth 1-forms locally.

Lemma 7.1.5 Let M be a smooth manifold and let ω be a smooth differential 1-form on M . Let $(U, \phi = (x^1, \dots, x^n))$ be a local chart on M . Then on the local chart,

$$\omega = \sum_{k=1}^n a_k dx^k$$

where $a_k \in C^\infty(M)$ are smooth functions for each k . Moreover, for each $p \in U$,

$$\omega_p = \sum_{k=1}^n a_k(p)(dx^k)_p$$

Proof For each point $p \in U$, the above proposition implies that

$$\omega_p = \sum_{k=1}^n a_k(p)(dx^k)|_p$$

By proposition 5.1.4, $a_k : U \rightarrow \mathbb{R}$ are smooth functions for each k . ■

Proposition 7.1.6 Let M be a smooth manifold. Let $f \in C^\infty(M)$ be a smooth function. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart on M . Then locally on the chart,

$$(df)_p = \sum_{k=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial x^k} \Big|_p (dx^k)_p$$

for $p \in U$.

Proof By the above lemma, we have

$$df = \sum_{k=1}^n a_k dx^k$$

for some smooth functions $a_k : U \rightarrow \mathbb{R}$. To compute a_k , we evaluate $\frac{\partial}{\partial x^j}$ on both sides to get $\frac{\partial f}{\partial x^j} = a_j$ and so we are done. ■

Notice that this coincides with the definition of a smooth vector field on local charts. Indeed this is no coincidence. To evaluate the total differential, simply plug in the value $\phi(p)$. We then obtain

$$(df)_p = \sum_{k=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial x^k} \Big|_{\phi(p)} (dx^k)|_p$$

Lemma 7.1.7 Let M be a smooth manifold. Let $f \in C^\infty(M)$ be a smooth function. Then $df : M \rightarrow T^*(M)$ is a smooth differential 1-form.

7.2 Differential k-Forms

Recall that $\Lambda(T^*(M))$ is a graded algebra that enjoys addition and wedging as multiplication.

Definition 7.2.1 (Differential k-Forms) Let M be a smooth manifold. A differential k -form on M is a section $s : M \rightarrow \Lambda^k(T^*(M))$ of exterior product of the cotangent bundle $p : T^*(M) \rightarrow M$. A smooth differential k -form on M is a smooth section of the vector bundle. We write $\Omega^k(M)$ for the vector space of all smooth differential k -forms on M . We write

$$\Omega^\bullet(M) = \bigoplus_{k=0}^n \Omega^k(M)$$

for the exterior algebra of smooth differential forms.

The direct sum is finite since we know that wedge products of $n + 1$ elements are necessarily zero in an n dimensional vector space.

Denote

$$I \in \{(i_1, \dots, i_k) \mid i_1, \dots, i_k \in \{1, \dots, n\}, i_1 < \dots < i_k\}$$

as a shorthand notation. Locally on a chart, we can rewrite

$$\omega = \sum_{1 \leq i_1 < \dots < i_k \leq n} a_{i_1, \dots, i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$$

in the following form. Moreover, under the following explicit formula, we have a characterization to whether ω is a smooth differential k -form. This is identical to the case for differential 1-forms.

Proposition 7.2.2 Let M be a smooth manifold and $(U, \phi = (x^1, \dots, x^n))$ a chart on M . Let ω be a differential k -form locally on the chart written as

$$\omega = \sum_I a_I dx^I$$

Then ω is a smooth differential k -form if and only if the coefficient functions $a_I : U \rightarrow \mathbb{R}$ are smooth functions.

Proposition 7.2.3 Let M be a smooth manifold. Let $\omega \in \Omega^k(M)$ and $\tau \in \Omega^l(M)$. Then $\omega \wedge \tau \in \Omega^{k+l}(M)$.

Proof We know that $\omega \wedge \tau \in \Omega^{k+l}(T(M))$ by construction of the exterior product. Explicitly: if $\omega = \sum f_I dx^I$ and $\tau = \sum g_J dx^J$, then

$$\omega \wedge \tau = \sum f_I g_J dx^I \wedge dx^J$$

Clearly if f_I and g_J are smooth, then their product is also smooth. ■

7.3 The Induced Pullback Map

Now let $f : M \rightarrow N$ be a smooth map of manifolds. Recall that for each $p \in M$, there is a pushforward map of tangent spaces: $(f_*)_p : T_p M \rightarrow T_{f(p)} N$. Taking dual of the tangent spaces, we obtain the dual map $(f^*)_p : T_{f(p)}^* N \rightarrow T_p^* M$. We can then take the exterior power of the map to obtain

$$\Lambda^k((f^*)_p) : \Lambda^k(T_{f(p)}^* N) \rightarrow \Lambda^k(T_p^* M)$$

which is the main object of the next definition. Denote this map by f^* .

Definition 7.3.1 (The Pullback Map) Let $f : M \rightarrow N$ be a smooth map of smooth manifolds. Define the pullback map of f on differential forms as follows. For $\omega \in \Omega^k(N)$, the map $f^* : \Omega^k(N) \rightarrow \Omega^k(M)$ is defined pointwise as

$$f^*(\omega_{f(p)}) = \Lambda^k((f^*)_p)(\omega_p)$$

A local expression would be better than the above abomination of a formula in this case.

Theorem 7.3.2 Let $f : M \rightarrow N$ be a smooth map of smooth manifolds. Let $(U, \phi = (x^1, \dots, x^m))$ be a chart on M and $(V, \psi = (y^1, \dots, y^n))$ be a chart on N such that $f(U) \subset V$.

- For dy^i an element of the dual basis of $T_{f(p)}^*(N)$, we have that

$$f^*(dy^i) = \sum_{k=1}^n \frac{\partial(f_i \circ \phi^{-1})}{\partial x_k} dx^k = df_i$$

- For $\tau = \sum_{k=1}^n c_k dy^k$ a smooth 1-form of N in local coordinates, we have that

$$f^*(\tau) = \sum_{k=1}^n (c_k \circ f) df_k$$

- For $\omega = \sum_I a_I dy^I$ a smooth k -form of N in local coordinates, we have that

$$f^*(\omega) = \sum_I (a_I \circ f) df_{i_1} \wedge \cdots \wedge df_{i_k}$$

- If f is a diffeomorphism, we have that

$$f^*(dy^1 \wedge \cdots \wedge dy^n) = \det \left(\frac{\partial \psi \circ f_i \circ \phi^{-1}}{\partial x_j} \right) dx^1 \wedge \cdots \wedge dx^n$$

Proposition 7.3.3 Let $f : M \rightarrow N$ and $g : N \rightarrow K$ be a smooth map of manifolds. Let $\omega_1, \omega_2, \omega \in \Omega^\bullet(M)$. The following properties hold for the pull back of forms.

- $\mathbb{R}$ -linearity: $f^*(a\omega_1 + b\omega_2) = af^*(\omega_1) + bf^*(\omega_2)$ for $a, b \in \mathbb{R}$
- Wedge product: $f^*(\omega_1 \wedge \omega_2) = f^*(\omega_1) \wedge f^*(\omega_2)$
- Respects composition: $(f \circ g)^*(\omega) = g^*(f^*(\omega))$

7.4 Exterior Derivatives

Definition 7.4.1 (Exterior Derivative) Let M be a smooth manifold. An exterior derivative on M is a map $d : \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$ such that the following are true.

- $d(\Omega^k(M)) \subseteq \Omega^{k+1}(M)$ is an $\mathbb{R}$ -linear map
- $d \circ d = 0$
- If $\omega \in \Omega^r(M)$ and $\tau \in \Omega^s(M)$ then $d(\omega \wedge \tau) = d\omega \wedge \tau + (-1)^r \omega \wedge d\tau$
- For any $f \in C^\infty(M)$, df is the differential of f as defined above.

Proposition 7.4.2 Let M be a smooth manifold. Then the exterior derivative exists and is unique. Moreover, if (U, ϕ) is a local chart on M and $\omega = \sum_I a_I dx^I \in \Omega^k(M)$ is a differential k -form, then locally on the chart

$$d\omega = \sum_I (da_I) \wedge dx^I = \sum_I \sum_{k=1}^n \frac{\partial a_I}{\partial x^k} dx^k \wedge dx^I$$

Proof Locally on each chart define $d : \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$ by the formula given. Then using a partition of unity subordinate to the open cover of M via the charts, patch it up globally. Now we check the above criteria for the exterior derivative.

- Differentiation in $\mathbb{R}^n$ is $\mathbb{R}$ -linear. The exterior derivative d is defined as differentiation on local charts of M so that d is also $\mathbb{R}$ -linear. By definition of the total differential, $d(\omega) \in \Omega^{k+1}(M)$ for $\omega \in \Omega^k(M)$.
- For $\omega \in \Omega^k(M)$, we have that

$$d^2\omega = \sum_I \sum_{j=1}^n \sum_{i=1}^n \frac{\partial^2 a_I}{\partial x^j \partial x^i} dx^i \wedge dx^j \wedge dx^I$$

Notice that for $i < j$, there are two copies of $\frac{\partial^2 a_I}{\partial x^j \partial x^i}$ in the triple sum since partial differentiation commutes while $dx^i \wedge dx^j = -dx^j \wedge dx^i$ by property of the exterior algebra. Thus the terms cancel out and we obtain $d^2\omega = 0$.

- It suffices to prove it for expressions of the form $\omega = f dx^I$ and $\tau = g dx^J$. By $\mathbb{R}$ -linearity this

extends to the rest of $\Omega^\bullet(M)$. We have that

$$\begin{aligned} d(\omega \wedge \tau) &= d(fg dx^I \wedge dx^J) \\ &= (f dg + g df) \wedge dx^I \wedge dx^J \\ &= (-1)^k f dx^I \wedge g \wedge dx^J + df \wedge dx^I \wedge g dx^J \\ &= d\omega \wedge \tau + (-1)^k \omega \wedge d\tau \end{aligned}$$

proving the claim.

- It is clear that f is a 0-form then df is the differential of f by the given equation.

It remains to show that the definition of d is invariant under the choice of chart so that the partition of unity can patch it up. Suppose that $(U, \phi = (x^1, \dots, x^n))$ and $(V, \psi = (y^1, \dots, y^n))$ are two charts on M such that

$$\omega = \sum_I f_I dx^I \quad \text{and} \quad \omega = \sum_I g_I dy^I$$

on (U, ϕ) and (V, ψ) respectively. Suppose that on (U, ϕ) we have the exterior derivative d and d' on (V, ψ) respectively. Then on $U \cap V$, we have

$$d\omega = d\left(\sum_I g_I dy^I\right) = \sum_I d(g_I) dy^I + \sum_I g_I d(dy^I)$$

But we have that

$$\begin{aligned} d(dy^I) &= d^2 y^I - dy^{i_1} \wedge d(dy^{i_2} \wedge \dots \wedge dy^{i_k}) \\ &= -dy^{i_1} \wedge d(dy^{i_2} \wedge \dots \wedge dy^{i_k}) \\ &\vdots \\ &= 0 \end{aligned}$$

so that $d\omega = \sum_I d(g_I) dy^I = d'\omega$ and we conclude. ■

Proposition 7.4.3 Let $f : M \rightarrow N$ be a smooth map of smooth manifolds. Let $\omega \in \Omega^k(N)$. Then

$$d(f^*\omega) = f^*(d\omega)$$

Proof It suffices to prove this on a local chart. Let $(V, \psi = (y^1, \dots, y^n))$ be a chart of N . We first prove the case for a 0-form. Let h be a 0-form. It suffices to show this for every p in the chart. Let $X_p \in T_p(N)$. Then on one hand, we have that

$$d(f^*h)_p(X_p) = X_p(f^*(h)) = X_p(h \circ f)$$

On the other hand, we have that

$$\begin{aligned} (f^*((dh)_p))(X_p) &= (dh)_{f(p)}(f^*(X_p)) && \text{(Pull back of a one form)} \\ &= (f^*(X_p))(h) && \text{(Definition of the differential)} \\ &= X_p(h \circ f) && \text{(Definition of } f^*) \end{aligned}$$

and so we are done.

Let ω be a smooth k -form. ω is given locally by $\omega = \sum_I a_I dy^I$ for some $a_I : V \rightarrow \mathbb{R}$ that is smooth. Then we have that $f^*(\omega) = \sum_I (a_I \circ f) df_{i_1} \wedge \dots \wedge df_{i_k}$ so that

$$d(f^*(\omega)) = \sum_I d(a_I \circ f) \wedge df_{i_1} \wedge \dots \wedge df_{i_k}$$

On the other hand, we have that

$$\begin{aligned} f^*(d\omega) &= f^* \left(\sum_I d(a_I) \wedge dy^I \right) \\ &= \sum_I d(a_I \circ f) \wedge df_{i_1} \wedge \dots \wedge df_{i_k} \end{aligned}$$

and so we are done. ■

Definition 7.4.4 (Closed and Exact Forms) Let α be a k -form of $\mathbb{R}^n$. Then α is closed if $d\alpha = 0$, and α is exact if there exists a $k-1$ -form such that $d\beta = \alpha$.

The reason for these names is because in algebraic topology, we can define the cohomology group of this chain of algebras (called a complex). In particular, it is defined by taking the closed k -forms and quotienting the exact k -forms.

7.5 Orientation of Smooth Manifold

Definition 7.5.1 (Orientation)

Let M be a smooth manifold of dimension n . An orientation of M is an equivalence class $[\omega] \in \frac{\Omega^n(M)}{\sim}$ of smooth n -forms where we say that $\omega_1 \sim \omega_2$ if there exists an everywhere positive smooth function $f \in C^\infty(M)$ such that $\omega_1 = f\omega_2$.

Definition 7.5.2 (Orientable Manifolds)

Let M be a smooth manifold. We say that M is orientable if M has an orientation.

Definition 7.5.3 (Oriented Atlas)

Let M be a smooth orientable manifold. Let $\{(U_i, \phi_i) \mid i \in I\}$ be an atlas on M . We say that the chart is oriented if for any two charts $(U, \phi = (x^1, \dots, x^n))$ and $(V, y^1, \dots, y^n)$ in the atlas we have that

$$\det \left(\left(\frac{\partial y^j \circ \phi^{-1}}{\partial x^i} \right)_{1 \leq i, j \leq n} \right) > 0$$

Proposition 7.5.4

Let M be a smooth manifold. Then M is orientable if and only if M admits an oriented atlas.

Proof Suppose that M is orientable. This means that we have a nowhere vanishing top form

$$\omega = f dx^1 \wedge \dots \wedge dx^n$$

in a coordinate chart $(U, \phi = (x^1, \dots, x^n))$ on M . By replacing $x^1 : U \rightarrow \mathbb{R}$ with $-x^1 : U \rightarrow \mathbb{R}$, we can assume that M is covered by such coordinate charts where the function corresponding to f above is positive in each chart. Suppose that $(V, \psi = (y^1, \dots, y^n))$ is another chart on M with non-empty intersection with U . Locally on V , suppose that ω takes the form of

$$\omega = g dy^1 \wedge \dots \wedge dy^n$$

Then on $U \cap V$, we have that

$$\begin{aligned} g dy^1 \wedge \dots \wedge dy^n &= (g \circ \phi^{-1}) \det \left(\frac{\partial(y^i \circ \phi^{-1})}{\partial x^j} \right) dx^1 \wedge \dots \wedge dx^n \\ &= f dx^1 \wedge \dots \wedge dx^n \end{aligned}$$

So if $f > 0$ and $g > 0$, we must have that the determinant greater than 0 and so we are done.

Conversely, suppose that the determinant is non zero on all intersections of a covering $\{U_\alpha \mid \alpha \in I\}$ of M . Choose a partition of unity $\{\theta_\alpha \mid \alpha \in I\}$ subordinate to the covering. Define a top form on

M by

$$\omega = \sum_{\alpha \in I} \theta_{\alpha} dy_{\alpha}^1 \wedge \cdots \wedge dy_{\alpha}^n$$

which is smooth since θ_{α} is smooth for each $\alpha \in I$. Let $(U_{\beta}, \psi = (y^1, \dots, y^n))$ be a coordinate chart in the covering. Then locally we have that

$$\omega|_{U_{\beta}} = \sum_{\alpha \in I} \theta_{\alpha} \det \left(\frac{\partial y_{\alpha}^i}{\partial x_{\beta}^j} \right) dy_{\beta}^1 \wedge \cdots \wedge dy_{\beta}^n$$

The form ω is non-negative since $\theta_{\alpha} > 0$ and the determinant is positive by assumption. It is non-zero since at each point, at least one of the functions θ_{α} is non-zero, as they all sum to 1. Thus we are done. ■

It is easy to see that a choice of orientation is a choice of coordinate charts such that either the nowhere vanishing form $\omega = f dx^1 \wedge \cdots \wedge dx^n$ has its coefficient f positive or negative throughout all coordinate charts. Because this way, the determinant of the transition functions will always have positive determinant.

Proposition 7.5.5 A connected manifold M has exactly two orientations.

Proposition 7.5.6 If a manifold with boundary M is orientable, then ∂M is also orientable.

Proof If $\dim(M) = 1$, then $\dim(\partial M) = 0$ and consists of points so that the result is trivial. Let $n = \dim(M) \geq 2$. Choose a local chart $(U, \phi = (x^1, \dots, x^n))$ such that ∂M is defined by $x_n = 0$ locally and on intersections with other charts $\det \left(\left(\frac{\partial y^i}{\partial x^j} \right)_{1 \leq i, j \leq n} \right) > 0$. Where two local charts with coordinates $(U, \phi = (x^1, \dots, x^n))$ and $(V, \psi = (y^1, \dots, y^n))$ intersect. Since $x_n = 0$ and $y^n = 0$, the Jacobian matrix for a point $p \in \partial M$ is given by

$$\begin{pmatrix} \frac{\partial y^1}{\partial x^1} \Big|_p & \cdots & \frac{\partial y^1}{\partial x^{n-1}} \Big|_p & \frac{\partial y^1}{\partial x^n} \Big|_p \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial y^{n-1}}{\partial x^1} \Big|_p & \cdots & \frac{\partial y^{n-1}}{\partial x^{n-1}} \Big|_p & \vdots \\ 0 & \cdots & 0 & \frac{\partial y^n}{\partial x^n} \Big|_p \end{pmatrix}$$

Now $\psi \circ \phi^{-1}$ maps $x_n > 0$ to $y_n > 0$. Thus if $x_n = 0$, then $y_n = 0$ and if $x_n > 0$, then $y_n > 0$. Thus

$$\frac{\partial y^n}{\partial x^n} \Big|_{x_n=0} > 0$$

and since the determinant of the above matrix is positive, the determinant of the upper left $(n-1) \times (n-1)$ matrix also has positive determinant. Thus $(U_{\alpha}|_{\partial M}, \phi|_{\alpha}|_{\partial M})$ give an atlas for ∂M whose transition functions have everywhere positive Jacobian. Thus ∂M is orientable. ■

In order to complete Stoke's theorem, we need a canonical orientation on the boundary on the manifold that depends on the choice of orientation on the manifold itself.

Definition 7.5.7 (Induced Orientation) Let M be a smooth orientable manifold with boundary of dimension n . Let ω be a nowhere vanishing 1-form representing an orientation on M . Suppose that in local coordinates, ω on the boundary of M is given by

$$\omega = \epsilon dx^1 \wedge \cdots \wedge dx^n$$

where $\epsilon = \pm 1$ and $x^n > 0$. Then define the induced orientation ω' on ∂M locally by

$$\omega' = (-1)^n \epsilon dx^1 \wedge \cdots \wedge dx^{n-1}$$

7.6 Integration of Smooth Differential k -forms

There are a number of conditions for integration on manifolds. Namely,

1. The manifold must be oriented
2. Only top forms can be integrated
3. The top forms must have compact support.

Once the above conditions are satisfied, we can use the partition of unity to define integration.

Definition 7.6.1 (Integration of Smooth Top Forms)

Let M be an oriented smooth manifold of dimension n . Let $\omega \in \Omega^n(M)$ be a smooth top form. Define the integral of ω over M as follows. Choose an oriented atlas $\{(U_i, \phi_i) \mid i \in I\}$ of M . Choose a partition of unity $\{\theta_i \mid i \in I\}$ subordinate to the open cover of M such that $\omega = \sum_{i \in I} \theta_i \omega$. Suppose that locally on each chart, we have $\theta_i \omega|_{U_i} = g_i dx_i^1 \wedge \cdots \wedge dx_i^n$ where each $g_i \in \mathcal{C}^\infty(M)$ has compact support. Then define the integral of ω over M by

$$\int_M \omega = \sum_{i \in I} \int_{\phi_i(U_i)} g_i \circ \phi_i^{-1}$$

where the integral on the right is the Riemann integral in $\mathbb{R}^n$.

One property of partition of unity is that only finitely many θ_α are not identically 0 on ω , so the above sum is finite. Moreover, since ω has compact support, each $g_\alpha \circ \phi_\alpha^{-1}$ is also of compact support. It remains to show that the definition of integral is independent of the choice of the partition of unity and the choice of the open cover.

Proposition 7.6.2 Let M be an orientable manifold of dimension n . Let $\omega \in \Omega^n(M)$ be a top form of compact support. Then $\int_M \omega$ is independent of the choice of the open cover and partition of unity.

Proposition 7.6.3 Let M be an oriented manifold of dimension n without boundary. Let $\omega \in \Omega^n(M)$ be a top form of M with compact support. If $-M$ denotes the opposite choice of orientation of M , then

$$\int_{-M} \omega = - \int_M \omega$$

We have actually seen examples of integration on manifolds before. Write $\omega = f_1 dx_1 + f_2 dx_2 + f_3 dx_3$ be a smooth top form in $\mathbb{R}^3$ that is compactly supported. Let $\gamma : I \rightarrow \mathbb{R}^3$ be a diffeomorphism. Then $C = \gamma(I)$ is a 1-dimensional manifold, and

$$\int_C \omega = \sum_{i=1}^3 \int_I f_i(\gamma(t)) \frac{d\gamma_i}{dt} dt$$

This is in fact the line integral of the vector field $f = (f_1, f_2, f_3)$ over C .

Theorem 7.6.4 Let M be an n -dimensional oriented manifold. Let ω be an $n - 1$ form with compact support. Then

$$\int_M d\omega = 0$$

Proof Choose a covering of M and a partition of unity $\{\theta_i \mid i \in I\}$ subordinate to the cover such

that we can write

$$\omega = \sum_{i \in I} \theta_i \omega$$

Then in a chart, we can write

$$\theta_i \omega = f_1 dx^2 \wedge \cdots \wedge dx^n - f_2 dx^1 \wedge dx^3 \wedge \cdots \wedge dx^n + \cdots + (-1)^{n+1} f_n dx^1 \wedge \cdots \wedge dx^{n-1}$$

so that

$$d(\theta_i \omega) = \left(\frac{\partial f_1}{\partial x_1} + \cdots + \frac{\partial f_n}{\partial x_n} \right) dx^1 \wedge \cdots \wedge dx^n$$

Now notice that

$$\int_{\mathbb{R}^n} \frac{\partial f_1}{\partial x_1} dx^1 \wedge \cdots \wedge dx^n = \int_{\mathbb{R}} \int_{\mathbb{R}} \cdots \left(\int_{\mathbb{R}} \frac{\partial f_1}{\partial x_1} dx^1 \right) dx^2 \cdots dx^n$$

using Fubini's theorem. Since f_1 has compact support, it vanishes for $|x_1| > N$ for some $N \in \mathbb{R}$. So

$$\int_{\mathbb{R}} \frac{\partial f_1}{\partial x_1} dx^1 = f_1(N) - f_1(-N) = 0$$

as N tends to infinity. Continuing in the same way, the entire integral evaluates to 0 and so we conclude. ■

This theorem in context says that integration on the manifold that does not touch the boundary contributes to the overall integration a sum of 0. Indeed we are assuming M to be an ordinary manifold without boundary in the above theorem.

Theorem 7.6.5 (Stokes' Theorem)

Let M be an n -dimensional oriented manifold with boundary ∂M carrying the induced orientation. Let ω be an $n-1$ form with compact support. Then

$$\int_M d\omega = \int_{\partial M} \omega|_{\partial M}$$

Proof Notice that when $n = 1$, it is entirely the same as the fundamental theorem of calculus so that we only treat the case for when $n \geq 2$. Write $\omega = \sum_{i \in I} \theta_i \omega$ where $\{\theta_i \mid i \in I\}$ is a partition of unity subordinate to a choice of open cover on M . Now we have that

$$\int_M d\omega = \sum_{i \in I} \int_M d(\theta_i \omega)$$

and we can write

$$\theta_i \omega = f_1 dx^2 \wedge \cdots \wedge dx^n - f_2 dx^1 \wedge dx^3 \wedge \cdots \wedge dx^n + \cdots + (-1)^{n+1} f_n dx^1 \wedge \cdots \wedge dx^{n-1}$$

similar to the above proof. There are two types of chart in the choice of open cover. If U_α does not intersect with the boundary, the above theorem implies that the contribution to the integral is 0. For those that do intersect, we have that

$$\begin{aligned} \int_M d(\theta_i \omega) &= \int_{x_n \geq 0} \left(\frac{\partial f_1}{\partial x_1} + \cdots + \frac{\partial f_n}{\partial x_n} \right) dx_1 \cdots dx_{n-1} \\ &= \int_{\mathbb{R}^{n-1}} [f_n]_0^\infty dx_1 \cdots dx_{n-1} \\ &= - \int_{\mathbb{R}^{n-1}} f_n(x_1, \dots, x_{n-1}, 0) dx_1 \cdots dx_{n-1} \\ &= \int_{\partial M} \theta_i \omega|_{\partial M} \end{aligned}$$

where the last equality follows because

$$\theta_i \omega|_{\partial M} = (-1)^{n-1} f_n dx^1 \wedge \cdots \wedge dx^{n-1}$$

and we used the induced orientation $(-1)^n dx^1 \wedge \cdots \wedge dx^{n-1}$. ■

As a result of Stoke's theorem, we prove several related theorems.

Example 7.6.6 (Divergence Theorem) Let S be a compact oriented and connected 2-dimensional manifold in $\mathbb{R}^3$ with smooth boundary. Let $F = (f_1, f_2, f_3)$ be a smooth vector field on S . Then

$$\int_S \operatorname{div}(F) dx_1 dx_2 dx_3 = \int_{\partial S} (F \cdot u) dA$$

where $u = \frac{n}{|n|}$ is the outward unit normal and $dA = |n| dx dy$.

Proof Let $(U, (x, y))$ be a chart on S . We will work with this chart until we patch things up locally. Suppose that on the chart, S is given by the graph of a smooth function $G : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then $h : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $(x, y) \mapsto (x, y, G(x, y))$ is a diffeomorphism. Write $\omega = f_1 dx_2 \wedge dx_3 + f_2 dx_3 \wedge dx_1 + f_3 dx_1 \wedge dx_2$. The pull back of ω by h is given by

$$\begin{aligned} h^*(\omega) &= f_1 \circ h \left(dy \wedge \left(\frac{\partial G}{\partial x} dx + \frac{\partial G}{\partial y} dy \right) \right) + f_2 \circ h \left(\left(\frac{\partial G}{\partial x} dx + \frac{\partial G}{\partial y} dy \right) \wedge dx \right) + f_3 \circ h(dx \wedge dy) \\ &= \left(-\frac{\partial G}{\partial x} f_1 - \frac{\partial G}{\partial y} f_2 + f_3 \right) dx \wedge dy \end{aligned}$$

Substituting into the integral, we have that locally on U , we have

$$\begin{aligned} \int_U \omega &= \iint_{\mathbb{R}^2} h^*(\omega) \\ &= \iint_{\mathbb{R}^2} \left(-\frac{\partial G}{\partial x} f_1 - \frac{\partial G}{\partial y} f_2 + f_3 \right) dx \wedge dy \\ &= \iint_{\mathbb{R}^2} F \cdot \begin{pmatrix} -\frac{\partial G}{\partial x} \\ -\frac{\partial G}{\partial y} \\ 1 \end{pmatrix} dx dy \end{aligned}$$

Locally on U , a tangent vector on S is a linear combination of $\frac{\partial h}{\partial x} = \begin{pmatrix} 1 \\ 0 \\ -\frac{\partial G}{\partial x_1} \end{pmatrix}$ and $\frac{\partial h}{\partial y} = \begin{pmatrix} 0 \\ 1 \\ -\frac{\partial G}{\partial x_2} \end{pmatrix}$

so that for any tangent vector $X_p = a_1 \begin{pmatrix} 1 \\ 0 \\ -\frac{\partial G}{\partial x_1} \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ -\frac{\partial G}{\partial x_2} \end{pmatrix}$, we have that

$$\begin{pmatrix} -\frac{\partial G}{\partial x} \\ -\frac{\partial G}{\partial y} \\ 1 \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_1 \frac{\partial G}{\partial x_1} + a_2 \frac{\partial G}{\partial x_2} \end{pmatrix} = 0$$

so that $n = \begin{pmatrix} -\frac{\partial G}{\partial x} \\ -\frac{\partial G}{\partial y} \\ 1 \end{pmatrix}$ is the normal vector to the level set G . Now define on U the two form $dA = |n| dx \wedge dy$. Then we have that

$$\iint_{\mathbb{R}^2} F \cdot \begin{pmatrix} -\frac{\partial G}{\partial x} \\ -\frac{\partial G}{\partial y} \\ 1 \end{pmatrix} dx dy = \iint_{\mathbb{R}^2} F \cdot u dA$$

where $u = \frac{n}{|n|}$. Since ω is compactly supported, choose a partition of unity subordinate to the compacy support of ω . Then we have that

$$\int_{\partial S} \omega = \int_{\partial S} F \cdot u dA$$

where the integral on the right is given the suitable partition function and that that partition can be chosen to be a finite cover so the sum of the partition functions and integral is finite since S is

compact. Now by Stoke's theorem, we have that

$$\int_{\partial S} \omega = \int_S d\omega = \int_S \operatorname{div}(F) dx_1 dx_2 dx_3$$

and so we conclude. ■

As one can see, often the hardest part of deducing all other identities in vector calculus is translating the problem in terms of differential forms into one that involves notions in vector calculus.

Example 7.6.7 (Green's Formula in the Plane) Let W be a compact and connected manifold with smooth boundary ∂W . Then

$$\int_{\partial W} (f dx + g dy) = \int_W \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$

Proof Set $\omega = f dx + g dy$, we then have by Stoke's theorem,

$$\begin{aligned} \int_{\partial W} \omega &= \int_W d\omega \\ &= \int_W d(f dx + g dy) \\ &= \int_W \left(-\frac{\partial f}{\partial y} dx \wedge dy + \frac{\partial g}{\partial x} dx \wedge dy \right) \\ &= \int_W \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy \end{aligned}$$

and so we conclude. ■

8 Vector Calculus in $\mathbb{R}^n$

8.1 The Gradient of a Scalar Function

Definition 8.1.1 (The Gradient of a Scalar Function)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $f : U \rightarrow \mathbb{R}$ be a differentiable function. Define the gradient of f to be function $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by

$$\text{grad}(f) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

Lemma 8.1.2

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $f : U \rightarrow \mathbb{R}$ be a differentiable function. Let $v \in \mathbb{R}^n$. Then we have

$$\text{grad}(f)(p) \cdot v = \partial_v f(p)$$

for any $p \in U$.

Definition 8.1.3 (Line Integrals)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $F : U \rightarrow \mathbb{R}^n$ be a function. Let $r : [a, b] \rightarrow U$ be a differentiable path. Let $C = \text{im}(r)$. Define the integral of F along r to be

$$\int_C F \, dr = \int_a^b F(r(t)) \cdot Jr \, dt$$

if the integral exists.

Theorem 8.1.4 (The Fundamental Theorem of Line Integrals) Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $f : U \rightarrow \mathbb{R}$ be a smooth function. Let $r : [a, b] \rightarrow U$ be a differentiable function in U . Let $C = \text{im}(r)$. Then we have

$$\int_C \text{grad}(f) \, dr = f(r(b)) - f(r(a))$$

Proof

Suppose that the components of r is given by $r(t) = (r_1(t), \dots, r_n(t))$. By definition of the line integral, we have that

$$\int_C \text{grad}(f) \, dr = \int_a^b \text{grad}(f)(r(t)) \cdot Jr \, dt = \int_a^b \sum_{i=1}^n \frac{\partial f}{\partial x_i} \Big|_{r(t)} \frac{dr_i}{dt} dt = \int_{[a,b]} r^*(\text{grad}(f))$$

Let $\omega_p = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \Big|_p dx_i$. Then ω is a one form on $\mathbb{R}^n$. We may pull back the one form along r onto the smooth manifold $\text{im}(r)$ to give the one form

$$r^*(\omega)_t = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \Big|_{r(t)} \frac{dr_i}{dt} dt$$

By definition of the integral of differential forms, we deduce that

$$\int_C \text{grad}(f) \, dr = \int_{[a,b]} r^*(\omega)$$

■

Definition 8.1.5 (Conservative Vector Fields)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $F : U \rightarrow \mathbb{R}^n$ be a function. We say that F is conservative if there exists a differentiable function $f : U \rightarrow \mathbb{R}$ such that $F = \nabla f$.

Proposition 8.1.6

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $F : U \rightarrow \mathbb{R}^n$ be a vector field. Then the following are equivalent.

- F is conservative.
- If $r, s : [a, b] \rightarrow U$ are differentiable paths in U such that $r(a) = s(a)$ and $r(b) = s(b)$, then we have

$$\int_{r([a,b])} f \cdot dr = \int_{s([a,b])} f \cdot ds$$

- If $r : [a, b] \rightarrow U$ is a differentiable path in U such that $r(a) = r(b)$, then we have

$$\int_{r([a,b])} f \cdot dr = 0$$

9 De Rham Cohomology

9.1 The Cohomology Groups

By proposition 6.2.4, the operator $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ for a smooth manifold M is in fact a boundary operator. Thus we can define a cochain complex using differential forms.

Definition 9.1.1 (De Rham Complex) Let M be a smooth manifold of dimension n . Define the de Rham complex of M to be the cochain complex

$$0 \longrightarrow \Omega^0(M) = C^\infty(M) \xrightarrow{d} \Omega^1(M) \longrightarrow \cdots \longrightarrow \Omega^{n-1}(M) \xrightarrow{d} \Omega^n(M) \longrightarrow 0$$

The complex is denoted as $(\Omega^\bullet(M), d)$, coinciding with the algebra of smooth differential forms.

Definition 9.1.2 (de Rham Cohomology Groups) Let M be a smooth manifold of dimension n . Define the de Rham cohomology groups of M to be the cohomology of the chain of differential forms:

$$H_{\text{dR}}^k(M) = H^k(\Omega^\bullet(M); \mathbb{R})$$

for $0 \leq k \leq n$.

Proposition 9.1.3 Let M be a smooth manifold of dimension n . Then the following are true for the de Rham cohomology of M .

- $H_{\text{dR}}^k(M)$ is a vector space over $\mathbb{R}$ for all $k \in \mathbb{N}$.
- For $r > n$ we have $H_{\text{dR}}^r(M) = 0$
- If M has k connected components then $H_{\text{dR}}^0(M) \cong \mathbb{R}^k$
- M is orientable if and only if $H_{\text{dR}}^n(M) \cong \mathbb{R}$. M is non-orientable if and only if $H_{\text{dR}}^n(M) = 0$.

Theorem 9.1.4

Let M be a smooth manifold of dimension n . Then the direct sum

$$H_{\text{dR}}^*(M) = \bigoplus_{i=0}^n H_{\text{dR}}^i(M)$$

is a graded $\mathbb{R}$ -algebra whose multiplication is given by exterior products.

Proposition 9.1.5 Let M, N be smooth manifolds and $f : M \rightarrow N$ a smooth map. Then f induces an $\mathbb{R}$ -linear map

$$f^* : H_{\text{dR}}^*(N) \rightarrow H_{\text{dR}}^*(M)$$

such that $f^*(a \wedge b) = f^*(a) \wedge f^*(b)$. Moreover, it is functorial:

- If $g : N \rightarrow K$ is another smooth map of manifolds, then $(g \circ f)^* = f^* \circ g^*$
- If $\text{id} : M \rightarrow M$ is the identity map on the manifold, then $\text{id}^* : H_{\text{dR}}^*(M) \rightarrow H_{\text{dR}}^*(M)$ is the trivial map on $\mathbb{R}$ -algebras.

Theorem 9.1.6 (Homotopy Invariance) Let $f : M \times I \rightarrow N$ be a smooth map of manifolds varying for each $t \in I = [0, 1]$. Write $f_t(x) = f(x, t)$. Then the pull back maps $f_0^*, f_1^* : H_{\text{dR}}^*(N) \rightarrow H_{\text{dR}}^*(M)$ are equal:

$$f_0^* = f_1^*$$

Theorem 9.1.7 If M is a compact, orientable connected n -dimensional manifold with a chosen orientation, then integration

$$\int_M : H_{\text{dR}}^n(M) \rightarrow \mathbb{R}$$

defines a canonical isomorphism between $H_{\text{dR}}^n(M)$ and $\mathbb{R}$.

9.2 De Rham's Theorem

Proposition 9.2.1 Let M be a smooth manifold. Then there is an isomorphism

$$\text{Hom}(H_k(M; \mathbb{Z}), \mathbb{R}) \cong H^k(M; \mathbb{R})$$

for any $k \in \mathbb{N}$.

Theorem 9.2.2 Let M be a smooth manifold. Define the map

$$I : H_{\text{dR}}^k(M) \rightarrow H^k(M; \mathbb{R})$$

as follows. For $[\omega] \in H_{\text{dR}}^k(M)$, $I([\omega]) \in H^k(M; \mathbb{R}) \cong \text{Hom}(H_k(M; \mathbb{Z}), \mathbb{R})$ sends

$$[c] \mapsto \int_c \omega$$

for $[c] \in H_k(M; \mathbb{Z})$. Then this map is an isomorphism so that

$$H_{\text{dR}}^k(M) \cong H^k(M; \mathbb{R})$$

for any $k \in \mathbb{N}$.

The isomorphism established is in fact even stronger. In particular, the two graded rings arising from the respective cohomology theories are isomorphic as graded rings and the exterior product corresponds to the cup product.

9.3 Computations of the Cohomology Groups

Proposition 9.3.1 The real space $\mathbb{R}^n$ has the de Rham cohomology

$$H_{\text{dR}}^k(\mathbb{R}^n) = \begin{cases} \mathbb{R} & \text{if } k = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proposition 9.3.2 The n -sphere S^n has the de Rham cohomology

$$H_{\text{dR}}^k(S^n) = \begin{cases} \mathbb{R} & \text{if } k = 0, n \\ 0 & \text{otherwise} \end{cases}$$

Theorem 9.3.3 Let $p, q \geq 1$, the sphere S^{p+q} is not diffeomorphic to any $M \times N$ manifolds where $\dim(M) = p$ and $\dim(N) = q$.

Proposition 9.3.4 Every smooth vector fields on S^{2n} vanishes at some point of the sphere.

Proposition 9.3.5 The real projective space $\mathbb{RP}^n$ has the de Rham cohomology

$$H_{\text{dR}}^k(\mathbb{RP}^n) = \begin{cases} \mathbb{R} & \text{if } k = 0 \text{ or } k = n \text{ where } n \text{ odd} \\ 0 & \text{otherwise} \end{cases}$$